Does Machine Learning Improve on Simple Tools for Osteoporosis Screening? Internal–External Validation, Calibration and Subgroup Performance in NHANES

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Abstract

Background. Machine learning (ML) models for osteoporosis screening are rarely compared with simple clinical tools under equal conditions.

Methods. We studied 9,657 adults aged ≥50 years in five NHANES cycles (2005–2018), not diagnosed with or treated for osteoporosis; the outcome was a hip T-score ≤ −2.5. Seven model families using 35 clinical, laboratory and socio-demographic predictors were tuned equally and recalibrated. In internal–external cross-validation (IECV) across cycles, logistic regression was compared with the Osteoporosis Self-Assessment Tool (OST), a sex-specific OST, ORAI, SCORE and an age–weight–sex model, and the other families with logistic regression. The main estimand, chosen after earlier analyses (all reported), was the pooled AUROC difference from OST; the equivalence margin was ±0.02. The study is exploratory.

Results. Penalised logistic regression was not outperformed: in IECV, five of six tuned ML families were equivalent to it, and none was better. In IECV it exceeded OST by 0.024 in AUROC (95% CI 0.009–0.038) and the age–weight–sex model by 0.019 (0.004–0.033); their clinical relevance was uncertain. A clinical model without blood tests, race or income exceeded OST by 0.016 (0.004–0.029). Referring 20% of people for DXA, the full model detected 2.8 (1.0–4.3) more cases per 1,000 screened than OST (8.6% more), and the clinical model 1.7 more.

Conclusions. Equally tuned ML did not outperform logistic regression. Thirty-five predictors gave a small, consistent gain over simple tools, about three cases per 1,000 screened at equal DXA capacity; whether that justifies a complex tool is a local judgement.

1 Introduction

Osteoporosis causes fragility fractures that lead to disability, loss of independence and excess mortality, yet effective treatment substantially lowers fracture risk once the disease is recognised [1]. Among US adults aged 50 years and older, about one in eight has osteoporosis at the femoral neck or lumbar spine [2]. Diagnosis relies on dual-energy X-ray absorptiometry (DXA), with osteoporosis defined as a T-score of −2.5 or lower [3]. DXA capacity is limited, and guidelines target screening mainly at women aged 65 and older and at younger postmenopausal women at increased risk, leaving men and many younger women under-assessed [4].

Simple risk tools can prioritise who most needs DXA. The Osteoporosis Self-Assessment Tool (OST), $0.2 \times (\text{weight} - \text{age})$, and its Asian counterpart OSTA use only age and weight [5, 6]; the Osteoporosis Risk Assessment Instrument (ORAI) and the Simple Calculated Osteoporosis Risk Estimation (SCORE) add a few questionnaire items [7, 8]. In women, these tools reach areas under the receiver operating characteristic curve (AUROC) of about 0.70–0.80 [5, 9, 10], and systematic reviews have found that adding further risk factors rarely improves on age and weight [11, 12]. OST also performs reasonably in men [13, 14, 15], although its accuracy varies with ethnicity [16]. FRAX, which estimates fracture probability, identifies low bone density less well than OST when calculated without BMD [17, 15], so it is not the natural comparator for a tool that selects people for DXA.

Machine learning (ML) can combine many routine predictors, and several groups have applied it to population surveys. In the Korea National Health and Nutrition Examination Survey (KNHANES), ML models reached AUROCs of 0.69–0.74 against 0.73 for logistic regression [18], and deep-learning models have also been proposed [19]. In the US National Health and Nutrition

Examination Survey (NHANES), recent models reported AUROCs of 0.81–0.85 [20, 21, 22]; one gradient-boosting model was compared with OST (0.83 vs 0.78) [21], and another study reported mainly accuracy [22]. These studies share features that make the added value of ML hard to judge: random splits from the same years, comparators that are absent or tuned less carefully than the ML models, accuracy as the main metric when most people do not have osteoporosis, cohorts that include people already diagnosed or treated, and little attention to calibration or to men. Across clinical prediction tasks, ML has rarely outperformed logistic regression [23], and class-imbalance corrections can distort calibration [24].

Our aim was not to propose another model, but to estimate how much ML and additional predictors add to simple tools for osteoporosis case-finding when every method is tuned equally and evaluated in the intended population and over time. The study adds to previous work in five ways: it is restricted to adults neither diagnosed with nor treated for osteoporosis; seven model families, logistic regression among them, are tuned with the same budget, the six others are tested for equivalence with logistic regression, and the four tree-based families are also combined in a stacking ensemble; all models are compared head-to-head with OST, a sex-specific OST, ORAI, SCORE and an age–weight–sex model in the same participants; performance is assessed by internal–external cross-validation across five survey cycles, with tuning nested inside each development set; and gains are expressed as cases detected per 1,000 people screened at a fixed DXA capacity, together with calibration, net benefit and performance in the strata targeted by screening guidelines. The study is reported according to TRIPOD+AI [25] (Supplementary Table S1).

2 Methods

2.1 Data source and study population

We used publicly available, de-identified data from NHANES, a cross-sectional survey of the non-institutionalised US civilian population with a complex, multistage probability design [26, 27]. We pooled the five 2-year cycles with femoral DXA: 2005–2006, 2007–2008, 2009–2010, 2013–2014 and 2017–2018, merging demographic, body-measure, DXA, laboratory, questionnaire and prescription-medication files on the respondent sequence number.

Of 50,463 participants, 14,166 were aged 50 years or older and 10,981 of these had a valid femoral neck bone mineral density (BMD) measurement. Because the intended use is case-finding before DXA, we excluded 1,324 participants who reported a physician diagnosis of osteoporosis (1,155) or current use of an osteoporosis medication without such a diagnosis (169); 282 of them met the outcome definition. The final cohort comprised 9,657 participants (Figure 1). No participant missed more than 50% of predictors. Participants without DXA are compared with the cohort in Supplementary Table S2, and the medication list is given in the Supplementary Methods.

2.2 Outcome

The outcome was osteoporosis at the hip, a T-score of -2.5 or lower at the femoral neck or total hip [3], calculated from the NHANES III reference values for non-Hispanic white women aged

20–29 years [28] for both sexes, as recommended by the International Society for Clinical Densitometry [29]. DXA was performed by trained technologists following the NHANES protocol, independently of the predictors.

The outcome was revised during the analysis. Earlier versions defined osteoporosis at the femoral neck or lumbar spine; we changed to the hip because the share of partial spine scans rose from 15.3% in 2005–2006 to 38.1% in 2017–2018, so a spine-based outcome would have changed definition between development and validation. Results of every version are reported in Supplementary Table S3, and two spine-inclusive outcomes were analysed as sensitivity analyses [30, 31].

Hip DXA was performed on Hologic QDR-4500A densitometers up to 2009–2010 and on Hologic Discovery densitometers thereafter, and we found no published hip cross-calibration between them. Mean femoral neck BMD in women was about 0.15 reference SD lower from 2013, coinciding with both the scanner change and the oversampling of non-Hispanic Asian adults. Hip-based prevalence was nevertheless stable (4.0%–4.8% by cycle). To test the effect of a possible scanner offset, we shifted the T-scores of one scanner period by 0.15 SD, redefined the outcome and refitted the models (Sensitivity analyses).

2.3 Predictors and data preparation

We considered 35 candidate predictors (32 variables, with race/ethnicity entered as four indicators) obtainable from history, examination or standard blood tests: age, sex, race/ethnicity (five categories), education and family income-to-poverty ratio (PIR); body mass index (BMI), weight, height and waist circumference; ten serum measurements (calcium, phosphorus, alkaline phosphatase, creatinine, albumin, uric acid, glucose, total protein, cholesterol and 25-hydroxyvitamin D); FRAX-type risk factors (prior hip, wrist or spine fracture, parental hip fracture, ever daily glucocorticoid use, current smoking, alcohol $\geq$3 drinks/day); and former smoking, postmenopausal status, ever use of female hormones, diabetes, arthritis, coronary heart disease, stroke and cancer. DXA-derived variables and self-reported osteoporosis were excluded to prevent leakage. Non-Hispanic Asian adults can be identified only from 2011, so in the models they fall within "other/multiracial"; this category made up 2.9%–4.6% of participants in 2005–2010 and 17.8% in 2017–2018.

Age (top-coded differently by cycle), creatinine (2005–2006 method change), vitamin D and alcohol (questionnaire change in 2017–2018) were harmonised across cycles as described in the Supplementary Methods. Reproductive variables were coded 0 for men. Missingness by cycle is shown in Supplementary Table S4.

2.4 Development and validation sets

Participants from 2005–2014 (n = 7,649; 343 cases) formed the development set and those from 2017–2018 (n = 2,008; 96 cases) the temporal validation set. In this temporal split, all fitting, tuning, threshold selection and recalibration used the development set only. The validation set had, however, been evaluated in earlier versions of the analysis, before the outcome and cohort were revised (Supplementary Table S3). No formal sample-size calculation was performed; the development set had 9.8 events per candidate predictor, so all regression models were penalised.

2.5 Missing data

Missing predictor values (at most 9.5%, for PIR) were imputed once by chained equations (scikit-learn IterativeImputer) [32], fitted within each training fold and applied to the corresponding held-out data; binary predictors were imputed on the continuous scale and not rounded. The outcome was not included in the imputation model. Single imputation understates the variance of the estimates slightly; missingness was below 10% for every predictor overall, although it reached 14.6% for vitamin D in 2007–2008 and 13.4% for PIR in 2017–2018 (Supplementary Table S4), so we expect the effect to be small.

2.6 Models and tuning

We compared eight model families: logistic regression (LR) with an L2 (ridge) penalty; LR with cubic B-splines for the 17 continuous or ordinal predictors, including 5-level education (LR-spline) [33]; a support vector machine (SVM); random forest (RF); XGBoost [34], LightGBM [35] and CatBoost [36]; and a stacking ensemble of the four tree-based models. Each was first fitted with default hyperparameters and three class-imbalance strategies (none, class weighting and SMOTE [37], applied to training folds only), using all 35 predictors and 5-fold stratified cross-validation. The seven single-learner families were then tuned with their best strategy by Bayesian optimisation (Optuna [38]), with 40 trials per family on the same folds (Supplementary Table S5); the stacking ensemble combined the tuned base learners and was not tuned further. The workflow is summarised in Supplementary Figure S1. The family with the highest tuned cross-validated AUROC was selected as the full model.

2.7 Benchmarks

The full model was compared on the same participants with: (i) OST [5], evaluated without fitting, including at its usual cut-offs (< 2 and < -1) [9]; (ii) a sex-specific OST, a logistic model of OST, sex and their interaction fitted on the development set, which allows the relation between OST and risk to differ in men; (iii) ORAI [7]; (iv) SCORE [8]; (v) LR with age, weight and sex; and (vi) a Boruta + XGBoost pipeline following Zhang et al. [22, 39]. ORAI and SCORE were developed in women and were applied to both sexes, with small adaptations to the available NHANES items (Supplementary Methods); they are also reported in women only. We also refitted the full model without race/ethnicity [40], as a sensitivity analysis of the full model rather than a competing tool. For calibration and net benefit, OST, ORAI and SCORE were converted to probabilities by logistic recalibration on the development set. A re-implementation of the published feature-selection analysis is reported in the Supplementary Methods.

2.8 Recalibration, thresholds and performance measures

Every model was recalibrated by Platt scaling [41] on its out-of-fold development predictions. The primary threshold was the development-set Youden index [42]; we also report a threshold giving 90% sensitivity in development and fixed risk thresholds. We assessed discrimination (AUROC and area under the precision–recall curve, AUPRC), calibration (calibration plots, slope, calibration-in-the-large and Brier score) [43] and clinical utility by decision curve analysis

[44]. Decision curves were restricted to thresholds of 2%–15%, the range in which a clinician would plausibly refer a patient for a low-cost, non-invasive scan.

Because AUROC differences are hard to interpret clinically [45], we also compared tools at fixed DXA capacity: for referral of 15%, 20%, 25% and 30% of people, ranked by each tool, we counted cases detected per 1,000 screened, the relative increase, and the number of people screened per additional case found. Unless stated otherwise, 95% confidence intervals (CIs) and P values came from 1,000 paired bootstrap resamples of the validation set; DeLong tests [46, 47] gave the same conclusions.

2.9 Estimands and tests

The primary estimand was the pooled IECV difference in AUROC between the full model and OST, with its 95% HKSJ CI (next subsection). We classified each gain as *relevant* if the lower 95% limit exceeded the ± 0.02 margin, *superior but of uncertain relevance* if the CI excluded zero but included 0.02, *equivalent* if the 90% CI lay within ± 0.02, and otherwise inconclusive. Because this estimand, the margin and the hierarchy below were fixed after earlier versions of the analysis had seen the data, the study as a whole is exploratory: the main estimand marks where we place most weight, not a prespecified confirmatory test. The 2017–2018 temporal split is the last IECV fold, not an independent confirmation; for it we report, descriptively, the hierarchy of three families used in earlier versions of this analysis, which was defined post hoc after the validation data had been examined:

1. **Primary:** the full model versus OST, the most widely used simple tool, at $\alpha = 0.05$.
2. **Secondary** (if the primary comparison was significant): the full model versus ORAI, SCORE, the three-variable model and the Boruta + XGBoost pipeline, with Holm adjustment [48] within this family of four.
3. **Equivalence of ML and regression:** each of the seven other families was considered equivalent to the full model if the 90% CI of the AUROC difference lay within ± 0.02 (two one-sided tests, each at $\alpha = 0.05$, without multiplicity adjustment). Because the full model was the family with the highest cross-validated AUROC, the test is slightly biased in favour of LR.

The margin was not taken from an external standard; it was chosen during the analysis as a difference we judged too small to matter. To apply it even-handedly, we also tested whether the full model was equivalent to OST and to the three-variable model, and we report the gains as cases detected so that readers can apply their own threshold [45]. All other comparisons, including those with the sex-specific OST and in subgroups, are exploratory, and no P value in this paper should be read as confirmatory.

2.10 Internal–external validation and model-fitting uncertainty

Because a single temporal split gives one estimate from 96 cases, we regard internal–external cross-validation (IECV) [49] as the main evidence on the size of the gain. Each of the five cycles was held out in turn and all models were developed on the other four. To keep the held-out cycle out of tuning, the penalty of each penalised regression model with more than three predictors was re-tuned within the four development cycles (40 Optuna trials, as in the main analysis);

the three-variable model used a fixed weak penalty ($C = 1$), which has negligible effect with three predictors and 7,493–8,235 participants per development set; each ML family likewise re-selected its imbalance strategy (with default hyperparameters) and was re-tuned with 40 Optuna trials within the four development cycles, so that no held-out cycle informed any model choice; the stacking ensemble was omitted for run time. Besides the full model, OST, the sex-specific OST and the three-variable model, we evaluated the full model without race/ethnicity; without race/ethnicity, income and education (clinical + laboratory); and without laboratory tests as well (clinical: 19 predictors from history and examination), because race, income and education are rarely recorded or contested and blood tests are not available for everyone. Several clinical items (parental hip fracture, glucocorticoid use, alcohol intake) are often missing or unreliable in routine records, so the clinical model approximates, but does not test, an EHR-based tool. AUROC differences within each cycle were estimated by paired bootstrap (CI as the difference ± 1.96 bootstrap SD) and pooled with a DerSimonian–Laird random-effects model [50], with Hartung–Knapp–Sidik–Jonkman (HKSJ) CIs, which are wider when there are few studies [51] (with the usual safeguard that the HKSJ variance is not smaller than the DerSimonian–Laird variance), and a 95% prediction interval; single AUROCs were pooled with Hanley–McNeil standard errors [52]. Because the development sets overlap, the cycles are not fully independent, and the pooled CIs are approximate; as a check, we also pooled the out-of-cycle predictions of all cycles and bootstrapped within cycles (one-stage estimate). Cases detected at fixed referral rates were computed on the pooled out-of-cycle predictions of all 439 cases, applying the referral rate within each cycle and breaking ties at random, with bootstrap CIs (resampling within cycles, conditional on the fitted models). We repeated this in the guideline strata, in the population not routinely screened (everyone except women aged 65 or older, since the USPSTF found insufficient evidence for screening men [4]), and with survey weights. To include uncertainty from model fitting in the temporal split, we also ran a refit bootstrap (1,000 replicates, hyperparameters fixed) in which both the development and validation sets were resampled and the models refitted.

2.11 Survey weighting and non-response

Models were fitted without survey weights. Pooled descriptive estimates used the examination weights divided by the number of cycles (five). For population-level performance in 2017–2018, we used the examination weights with design-based CIs from a Rao–Wu rescaled bootstrap over primary sampling units within strata (500 replicates) [53, 54]. Because 22.5% of eligible participants had no valid DXA, we also multiplied the examination weights by the inverse of the modelled probability of having a DXA result (logistic regression on age, sex, education, PIR, BMI, race/ethnicity and cycle). We report the distribution of these weights and, because large weights can make estimates unstable, repeated the analysis with weights trimmed at the 99th percentile, at five times the median, and with the inverse-probability factor capped at 3 [55, 56].

2.12 Subgroups, clinical strata and sensitivity analyses

In the validation set we evaluated performance by sex, age group and race/ethnicity, reporting non-Hispanic Asian adults separately, and in four strata that correspond to screening recommendations [4]: women aged 50–64 and $\geq$65 years, and men aged 50–69 and $\geq$70 years. We

also evaluated sex-specific Youden thresholds. Sensitivity analyses repeated development and validation for the two spine-inclusive outcomes, with diagnosed or treated participants included, for the two scanner-shift scenarios, without non-Hispanic Asian participants, on 2013–2014 alone (same scanner type as 2017–2018), without 2005–2006, without laboratory predictors and without alcohol. Predictor importance (SHAP) [57] and a 10-predictor short form without blood tests are reported in the Supplementary Methods and Results.

2.13 Software

Analyses used Python 3.9.6 with scikit-learn 1.6.1 [58], XGBoost 2.1.4, LightGBM 4.6.0, CatBoost 1.2.10, Optuna 5.0.0, SHAP 0.49.1 and statsmodels 0.14.6 (full list in the code repository), with a fixed random seed. Two-sided $P < 0.05$ was considered significant.

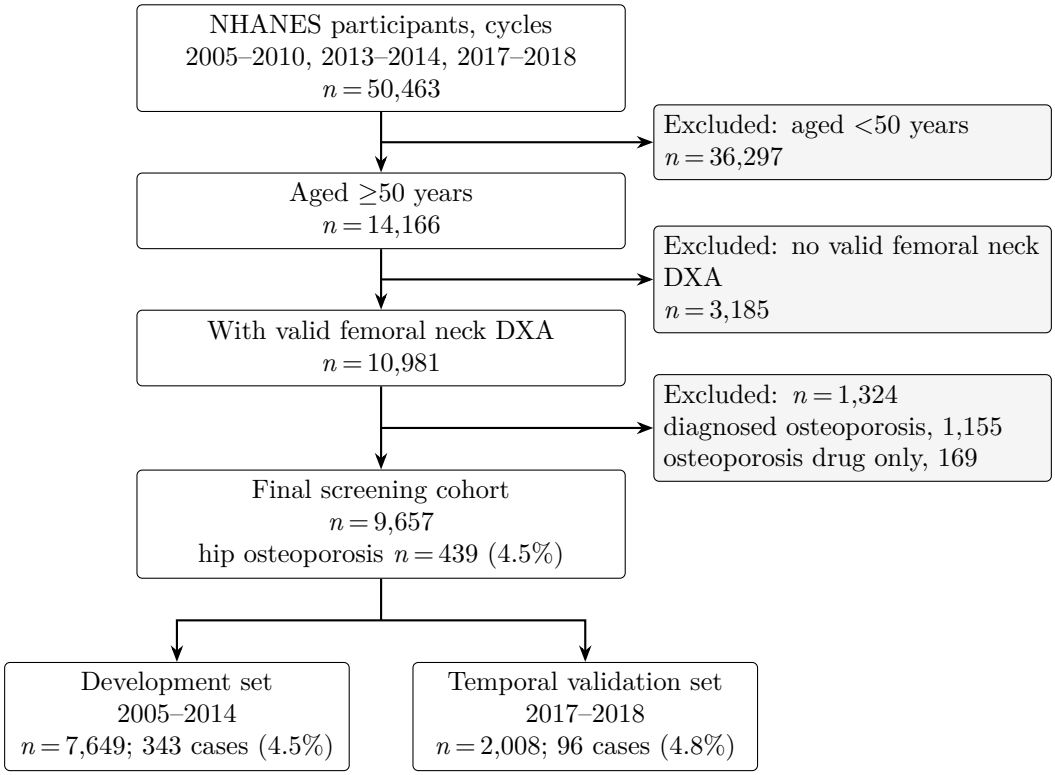


Figure 1: Participant flow. Percentages are the prevalence of osteoporosis at the hip (femoral neck or total hip T-score ≤ -2.5). No participant was excluded for missing more than 50% of predictors.

3 Results

3.1 Participants

The cohort comprised 9,657 adults (mean age 64.2 years; 43.0% women), of whom 439 (4.5%) had undiagnosed osteoporosis at the hip (survey-weighted 3.9%). Prevalence was 4.0%–4.8% by cycle: 4.5% in the development set and 4.8% in the validation set. Participants with osteoporosis were older, more often women and postmenopausal, smaller and poorer, and more often had a prior fracture (22.3% vs 11.6%) (Table 1). The validation cohort had higher vitamin D and alkaline phosphatase, lower albumin, more diabetes, less hormone use and more participants

of other or multiracial ethnicity (17.8% vs 5.9%) than the development cohort (Supplementary Table S6).

Table 1: Baseline characteristics by osteoporosis status (n = 9,657). Values are mean (SD) or n (%); weighted columns are survey-weighted means or percentages. P values are from unweighted Welch's t-tests or χ^2 tests. Percentages are of participants with a non-missing value; reproductive variables were coded 0 for men, so their percentages include men in the denominator.

	Unweighted			Weighted	
Characteristic	No (n = 9,218)	Yes (n = 439)	P	No	Yes
Age, years	63.88 (9.25)	71.14 (9.07)	<0.001	61.9	70.0
Female sex	3,833 (41.6)	318 (72.4)	<0.001	45.3	76.5
Postmenopausal	3,351 (37.1)	293 (70.4)	<0.001	39.7	74.7
Ever used female hormones	1,299 (14.4)	70 (16.9)	0.18	17.8	19.5
Prior hip, wrist or spine fracture	1,069 (11.6)	98 (22.3)	<0.001	13.4	23.4
Parental hip fracture	889 (9.9)	45 (10.7)	0.65	11.1	12.5
Ever daily glucocorticoid use	495 (5.4)	22 (5.1)	0.85	6.1	7.3
Alcohol ≥3 drinks/day	222 (2.5)	3 (0.7)	0.036	2.5	1.3
Current smoker	1,562 (17.0)	92 (21.0)	0.035	15.6	25.3
Stroke	512 (5.6)	45 (10.3)	<0.001	4.4	8.4
Diabetes	1,771 (19.2)	77 (17.5)	0.42	14.7	12.3
Non-Hispanic Black	1,991 (21.6)	43 (9.8)	<0.001	9.7	4.6
Income-to-poverty ratio	2.74 (1.62)	2.10 (1.37)	<0.001	3.33	2.43
BMI, kg/m^2	28.94 (5.53)	24.41 (4.46)	<0.001	28.97	24.37
Weight, kg	81.51 (17.75)	62.23 (13.56)	<0.001	82.8	62.4
Height, cm	167.67 (9.84)	159.44 (8.93)	<0.001	168.8	159.9
Waist, cm	101.60 (13.62)	90.73 (13.29)	<0.001	101.7	90.0
Alkaline phosphatase, U/L	74.31 (24.62)	82.85 (28.38)	<0.001	71.9	82.5
Uric acid, mg/dL	5.72 (1.43)	5.29 (1.44)	<0.001	5.62	5.12
25(OH) vitamin D, nmol/L	67.37 (27.85)	67.17 (30.44)	0.90	72.8	69.3

3.2 Model selection

With default hyperparameters, cross-validated AUROC ranged from 0.785 (SVM) to 0.876 (LR) (Supplementary Table S7). Equal tuning narrowed the gaps: SVM gained 0.035, LightGBM 0.023 and XGBoost 0.018, but LR only 0.002, and seven of eight families then lay within 0.006 of each other (0.873–0.879; RF 0.864) (Supplementary Table S8). Penalised LR remained the highest (0.879) and was selected as the full model. Tuned XGBoost and LightGBM converged on stumps (depth 1; two leaves), that is, on additive models, and CatBoost on shallow trees (depth 4). Class weighting and SMOTE raised the raw Brier score of LR from 0.035 to 0.127–0.137; recalibration removed this difference.

3.3 Internal–external validation across cycles

With the penalty re-tuned within each set of development cycles, the full model exceeded both OST and the three-variable model in every held-out cycle, although only by 0.002 over the three-variable model in 2005–2006 (Table 2). The pooled gain was 0.024 (95% HKSJ CI 0.009–0.038; prediction interval 0.007–0.040) over OST and 0.019 (0.004–0.033) over the three-variable model, with I^2 of 0% and 29%. The full model was therefore better than both, but whether the gain exceeds the ±0.02 margin is uncertain: the CIs include values both below and above 0.02, so the gain is neither equivalent nor demonstrably relevant. The gain over the sex-specific OST was

0.017 (0.004–0.029), and the sex-specific OST exceeded OST by 0.006 (0.001–0.012). Without race/ethnicity alone, the full model had a pooled AUROC of 0.877 and exceeded OST by 0.021 (0.007–0.034) and the three-variable model by 0.016 (−0.001 to 0.033). The clinical + laboratory model, without race, income and education, retained most of the gain: 0.020 (0.007–0.033) over OST and 0.016 (−0.0005 to 0.032) over the three-variable model. The clinical model, which also omits blood tests, had a pooled AUROC of 0.871 (0.831–0.910) and exceeded OST by 0.016 (0.004–0.029) and the three-variable model by 0.011 (−0.005 to 0.027); laboratory tests added 0.005 (0.000–0.010) to it, and the full model exceeded it by 0.011 (0.003–0.019) (Supplementary Table S9). Its penalty was nearly unpenalised when 2005–2006 was held out ($C = 31.7$), as was the full model's (below). The full model's AUROC ranged from 0.841 (2013–2014) to 0.912 (2005–2006); calibration slopes ranged from 0.82 to 1.25 and the ratio of observed to expected cases from 0.84 to 1.16. Re-tuning changed the pooled AUROC by only 0.001 compared with using the penalty tuned on 2005–2014 (0.881 vs 0.882; the latter leaks four held-out cycles into tuning). Pooling the out-of-cycle predictions instead of meta-analysing per-cycle differences gave a gain over OST of 0.026 (0.015–0.036) (Supplementary Table S10). The selected penalty was stable ($C = 0.13$–0.94) except when 2005–2006 was held out ($C = 22.5$, nearly unpenalised), which made little difference to that fold's AUROC (0.912 vs 0.917).

With every ML family re-tuned inside each set of development cycles, SVM, LR-spline, XGBoost, LightGBM and CatBoost were equivalent to the full model (pooled differences −0.006 to −0.002; all 90% CIs within −0.012 to 0.005), and none was better; RF was not equivalent (−0.010; 90% CI −0.021 to 0.000; 95% CI −0.024 to 0.004) (Supplementary Table S11). Nesting changed the pooled differences by at most 0.003 compared with hyperparameters fixed at their 2005–2014 values.

Table 2: Internal–external cross-validation: each cycle held out in turn, with models developed on the other four cycles and the penalty of each regression model except the three-variable model (fixed $C = 1$) re-tuned within the development cycles (40 Optuna trials). Δ, full model minus comparator (paired bootstrap within the held-out cycle; 95% CI as difference ±1.96 bootstrap SD). Pooled estimates from a DerSimonian–Laird random-effects model with Hartung–Knapp–Sidik–Jonkman (HKSJ) 95% CIs; single AUROCs pooled with Hanley–McNeil standard errors. O/E, observed to expected cases. R/S-free, full model without race/ethnicity, income and education; AWS, age–weight–sex model; OST-sex, sex-specific OST.

			AUROC							
Held-out cycle	n	Cases	Full	R/S-free	AWS	OST-sex	OST	Δ vs OST (95% CI)	Δ vs AWS (95% CI)	Slope; O/E
2005–2006	1,422	65	0.912	0.906	0.910	0.909	0.900	0.012 (−0.011 to 0.036)	0.002 (−0.014 to 0.019)	1.09; 0.88
2007–2008	2,090	84	0.901	0.902	0.873	0.873	0.867	0.034 (0.009–0.058)	0.028 (0.006–0.050)	1.25; 0.84
2009–2010	2,164	100	0.877	0.874	0.853	0.858	0.858	0.019 (0.002–0.036)	0.024 (0.007–0.041)	0.93; 1.08
2013–2014	1,973	94	0.841	0.828	0.811	0.813	0.808	0.033 (0.004–0.062)	0.030 (0.004–0.056)	0.82; 1.16
2017–2018	2,008	96	0.869	0.863	0.851	0.850	0.841	0.028 (0.005–0.051)	0.018 (−0.003 to 0.038)	1.00; 1.07
Pooled	9,657	439	0.881	0.876	0.861	0.862	0.856	0.024 (0.009–0.038)	0.019 (0.004–0.033)	
95% CI			0.847–0.915	0.837–0.914	0.817–0.905	0.819–0.905	0.815–0.897			
90% CI (±0.02)								0.013–0.035	0.007–0.030	

3.4 Temporal validation

The 2017–2018 cycle, the last IECV fold, is reported in full because earlier versions of the analysis used it as the validation set. There the full model achieved an AUROC of 0.869 (95% CI 0.831–0.904), close to its cross-fitted internal estimate (0.877; the tuning estimate of 0.879

used the tuning folds), with a calibration slope of 1.00 (0.85–1.18) (Table 3, Figure 2). The P values below are descriptive.

Primary comparison. The full model exceeded OST (0.841) by 0.028 (0.007–0.052; $P = 0.014$). When the development set was also resampled and the models refitted (1,000 replicates), the CI widened to −0.003 to 0.052 (difference ≤ 0 in 4.4% of replicates); for the three-variable model it was −0.011 to 0.038. In the temporal split alone, the 90% CIs of the gains over OST (0.010–0.048) and over the three-variable model (0.002–0.036) also extended beyond the equivalence margin.

Secondary comparisons. After Holm adjustment, the full model exceeded ORAI (0.791; difference 0.078, 0.047–0.115; $P_{\text{Holm}} = 0.004$) but not SCORE (0.841; 0.029, 0.003–0.055; $P_{\text{Holm}} = 0.084$), the three-variable model (0.851; 0.018, −0.001 to 0.039; $P_{\text{Holm}} = 0.14$) or the Boruta + XGBoost pipeline (0.856; 0.013, −0.007 to 0.030; $P_{\text{Holm}} = 0.18$).

ML versus regression. Tuned ML families had AUROCs of 0.862 (RF) to 0.872 (stacking). Six of seven were equivalent to LR within ±0.02. For RF, the upper 90% limit (0.0204) exceeded the margin, so equivalence could not be concluded, although RF was not better (difference 0.007). SVM had a slightly higher AUPRC (0.367 vs 0.335) and lower Brier score (0.037 vs 0.038) than LR.

Sex-specific OST. Allowing the relation between OST and risk to differ by sex raised OST's AUROC from 0.841 to 0.850 (difference 0.009, 0.002–0.017), similar to the three-variable model (0.851). Within each sex, the sex-specific OST ranks people exactly as OST does; the gain comes from placing men and women on a common risk scale. The full model exceeded the sex-specific OST by 0.019 (−0.0004 to 0.041; $P = 0.060$; exploratory) (Supplementary Table S12).

Where the gain comes from. In nested models, adding other body measures, income and education to age, weight and sex raised the AUROC by 0.007, FRAX-type clinical risk factors and history by 0.001, laboratory tests by 0.005 and race/ethnicity by 0.006 (Supplementary Table S13). The full model without race/ethnicity had an AUROC of 0.863, 0.006 (−0.001 to 0.012) lower than the full model.

3.5 Cases detected at fixed DXA capacity

Across the five held-out cycles (45.5 cases per 1,000 screened), the full model detected more cases than every simple tool at each referral rate (Table 4). If 20% of people were referred for DXA (200 scans per 1,000 screened), it would detect 35.3 cases per 1,000, against 35.0 for the clinical + laboratory model, 34.2 for the clinical model, 33.6 for the sex-specific OST, 33.1 for the three-variable model and 32.5 for OST. Its advantage over OST was 2.3–3.0 cases per 1,000 at referral rates of 15%–30% (6.9%–8.6% more cases), all with CIs excluding zero, so that about 330–430 people would need to be screened with the full model rather than OST to find one additional case with the same number of scans. The clinical model captured about 60% of this gain (1.7 of 2.8 cases per 1,000 at 20% referral). In the population not routinely screened (216 cases), the gain over OST was 0.029 (0.012–0.046) in AUROC and 2.3 (0.9–3.9) cases per 1,000 at 20% referral, 12% more cases. In all participants with survey weights, the gain was 2.4 (0.5–4.5) per 1,000 (Supplementary Table S10). Breaking ties at random rather than by order changed these counts by at most 0.1 per 1,000. Its advantage over the sex-specific OST was

Table 3: Temporal validation performance (n = 2,008; 96 with osteoporosis), with 95% bootstrap CIs. For OST, ORAI and SCORE, Brier score and slope are for the recalibrated score. ΔAUROC, full model minus each alternative (paired bootstrap; 95% percentile CI, except 90% CI for the equivalence family); P, two-sided bootstrap P; P_{Holm}, Holm-adjusted within its family; not shown for the equivalence family, whose conclusion rests on the 90% CI. The hierarchy was defined post hoc. Sensitivity, specificity and referrals of every model are in Supplementary Table S14.

Model	AUROC (95% CI)	AUPRC	Brier	Slope (95% CI)	ΔAUROC (95% CI)	P	P_{Holm}
Full model (LR, 35 predictors)	**0.869 (0.831–0.904)**	**0.335**	**0.038**	**1.00 (0.85–1.18)**	–	–	–
Primary comparison							
OST	0.841 (0.796–0.883)	0.280	0.040	0.98 (0.79–1.20)	0.028 (0.007–0.052)	0.014	–
Secondary comparisons (Holm-adjusted, family of four)							
LR: age, weight, sex	0.851 (0.808–0.890)	0.312	0.039	1.04 (0.86–1.28)	0.018 (−0.001 to 0.039)	0.070	0.14
SCORE	0.841 (0.799–0.879)	0.263	0.040	1.04 (0.87–1.25)	0.029 (0.003–0.055)	0.028	0.084
ORAI	0.791 (0.742–0.834)	0.177	0.042	0.93 (0.75–1.11)	0.078 (0.047–0.115)	0.001	0.004
Boruta + XGBoost pipeline	0.856 (0.820–0.892)	0.295	0.039	1.04 (0.89–1.24)	0.013 (−0.007 to 0.030)	0.18	0.18
Tuned ML families vs LR: equivalence (90% CI, margin ±0.02)							
LR-spline	0.869 (0.832–0.903)	0.320	0.038	1.06 (0.91–1.25)	0.000 (−0.008 to 0.007)	0.94	–
SVM	0.870 (0.832–0.906)	0.367	0.037	1.03 (0.87–1.22)	−0.001 (−0.010 to 0.009)	0.87	–
Random forest	0.862 (0.823–0.900)	0.313	0.038	1.04 (0.87–1.24)	0.007 (−0.008 to 0.020)	0.40	–
XGBoost	0.868 (0.834–0.901)	0.320	0.038	1.04 (0.90–1.22)	0.001 (−0.010 to 0.012)	0.88	–
LightGBM	0.867 (0.832–0.902)	0.325	0.038	1.06 (0.92–1.26)	0.002 (−0.010 to 0.012)	0.83	–
CatBoost	0.867 (0.830–0.905)	0.333	0.038	1.07 (0.92–1.26)	0.002 (−0.011 to 0.013)	0.86	–
Stacking	0.872 (0.836–0.908)	0.338	0.038	1.04 (0.89–1.24)	−0.003 (−0.012 to 0.006)	0.51	–
Exploratory							
OST, sex-specific	0.850 (0.806–0.890)	–	–	–	0.019 (−0.0004 to 0.041)	0.060	–
Full model without race	0.863 (0.825–0.898)	0.325	0.038	1.00 (0.84–1.19)	0.006 (−0.001 to 0.012)	0.14	–

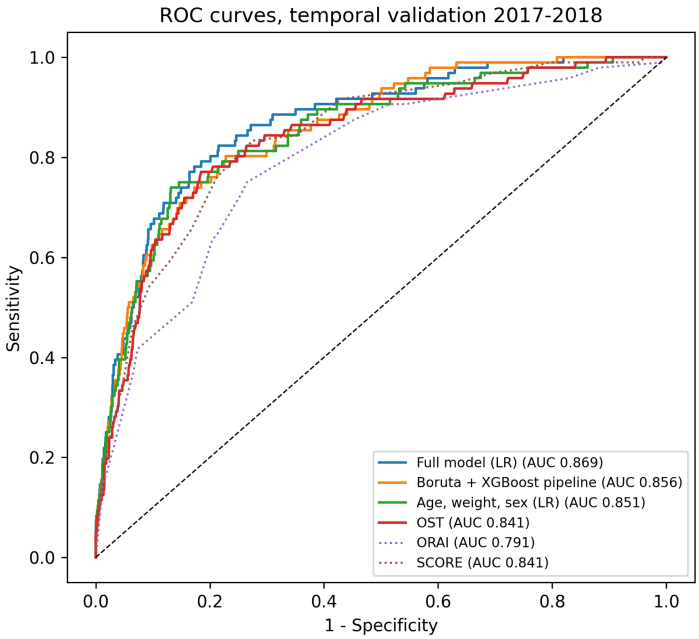


Figure 2: Receiver operating characteristic curves in the temporal validation set (2017–2018) for the full model and the simple or published benchmarks (ORAI and SCORE dotted).

1.8–2.4 per 1,000, and the sex-specific OST itself gained −0.1 to 1.0 per 1,000 over OST. In the 2017–2018 split alone, the advantage over OST was 2.0–3.0 per 1,000, with CIs including zero; at 20% referral, SCORE detected 31.9 and ORAI 24.9 cases per 1,000 (Supplementary Table S15). At OST's published cut-offs, the full model at the same referral rate detected 3 (−2 to 8) more of the 96 cases than OST < 2 (39.1% referred) and 6 (−2 to 14) more than OST < -1 (14.9% referred) (Supplementary Table S16).

Table 4: Cases detected per 1,000 people screened when a fixed proportion, ranked by each tool, is referred for DXA, from the pooled out-of-cycle predictions of the internal–external cross-validation (9,657 participants; 439 cases, 45.5 per 1,000). Referral was applied within each cycle; ties were broken at random (estimates averaged over 200 draws). Δ, full model minus comparator, with 95% CIs from 1,000 bootstrap resamples within cycles (conditional on the fitted models); % more, relative increase; per extra case, people screened per additional case found (1,000/Δ) with the same number of scans. Clinical + lab: without race/ethnicity, income and education. Clinical: also without laboratory tests (19 predictors).

	Cases detected per 1,000 screened						Full model vs OST		Full model vs clinical	
DXA scans per 1,000	Full	Clinical + lab	Clinical	Age, weight, sex	OST, sex-specific	OST	Δ (95% CI); % more	Per extra case	Δ (95% CI)	Per extra case
150	31.8	31.5	30.2	29.2	29.4	29.5	2.3 (0.5–4.0); 7.9%	432	1.6 (0.0–2.9)	644
200	35.3	35.0	34.2	33.1	33.6	32.5	2.8 (1.0–4.3); 8.6%	358	1.1 (−0.2 to 2.4)	878
250	38.3	37.3	36.3	36.0	36.1	35.3	3.0 (1.4–4.7); 8.5%	333	2.0 (0.7–3.0)	508
300	40.0	39.2	38.9	37.8	38.0	37.4	2.6 (0.9–3.6); 6.9%	386	1.0 (0.2–2.1)	966

3.6 Calibration and clinical utility

The full model was well calibrated internally (cross-fitted slope 0.99) and in temporal validation (Brier score 0.038) (Figure 3). It slightly underestimated risk in 2017–2018 (calibration-in-the-large 0.09, −0.16 to 0.29), as did the recalibrated simple tools (0.13–0.16); none of these intercepts differed significantly from 0. Across thresholds of 2%–15%, every model, including ORAI, had higher net benefit than referring everyone or no one (Figure 4). ORAI had the lowest net benefit throughout, and the differences between the other models were small relative to their uncertainty (Supplementary Table S17); at a threshold of 15%, for example, the Boruta + XGBoost re-implementation had slightly higher net benefit than the full model (0.015 vs 0.012). At a threshold of 10%, net benefit was 0.022 (0.015–0.030) for the full model and 0.018 (0.011–0.026) for OST, that is, 2.2 versus 1.8 net true positives per 100 people screened (3.1 vs 2.8 true positives).

Weighted to the 2017–2018 population, AUROCs were slightly higher and ranked the models similarly (full model 0.889, three-variable model 0.870, OST 0.868; Supplementary Table S18). With non-response adjustment, the full model's AUROC was 0.872 (0.812–0.934) and its difference from OST 0.023 (−0.003 to 0.051). The weighted observed prevalence was 4.2%, and the mean predicted risk 4.1% with examination weights and 3.9% with non-response-adjusted weights. The non-response factor had a median of 1.17 and a 99th percentile of 3.85, but reached 32.5, with 11 participants above 10. Trimming or capping the weights gave AUROCs of 0.870–0.888 and gains over OST of 0.021–0.025 (Supplementary Table S19). Trimming at five times the median raised the weighted prevalence from 4.2% to 4.7%, showing that the few heavily weighted participants had lower than average prevalence; the ranking of the models did not change.

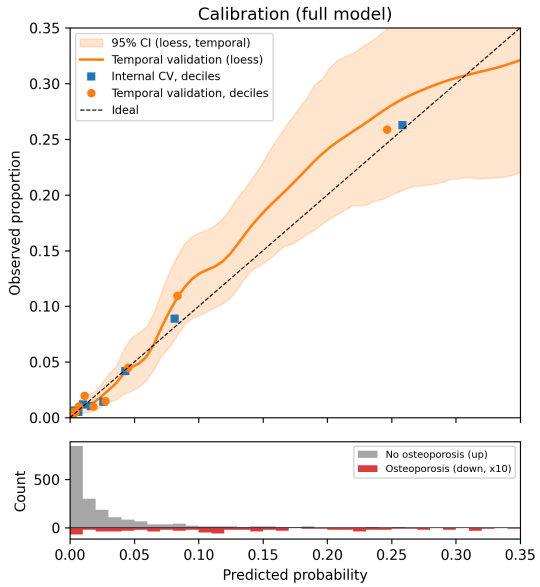


Figure 3: Calibration of the full model: deciles of predicted risk internally (cross-fitted) and in temporal validation, and a loess curve for the validation set with a 95% bootstrap band (200 replicates); dashed line, perfect calibration. The axes stop at 0.35 (98th percentile of predictions). Lower panel: predicted risks of participants without (up) and with (down, counts ×10) osteoporosis.

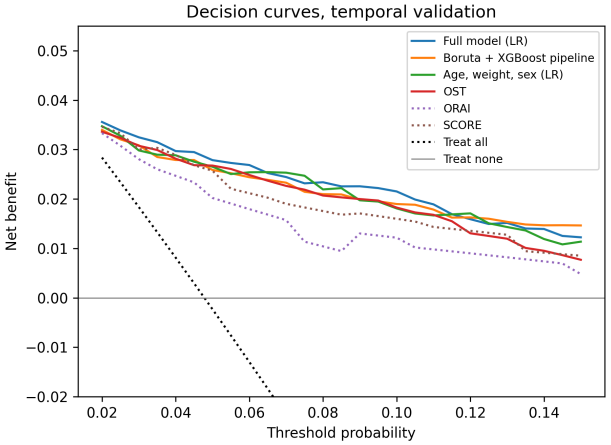


Figure 4: Decision curves in the temporal validation set over threshold probabilities of 2%–15% (OST, ORAI and SCORE recalibrated to probabilities).

3.7 Subgroups and clinical strata

At the global threshold, the full model detected 61 of 72 cases in women (sensitivity 0.847) but 16 of 24 in men (0.667), although it discriminated similarly within each sex (AUROC 0.853 and 0.820) (Supplementary Table S20). Sex-specific thresholds (6.7% for women, 3.5% for men) detected the same 77 cases with fewer referrals (22.8% vs 23.8%), raising sensitivity in men to 0.750. In women only, the full model exceeded ORAI (AUROC 0.802; difference 0.051, 0.021–0.086) but not SCORE (0.843; 0.011, −0.020 to 0.042) or OST (0.834; 0.019, −0.005 to 0.046) (Supplementary Table S12). Non-Hispanic Asian adults had a high prevalence (7.3%) and the model discriminated well in them (AUROC 0.904); other racial and ethnic subgroups had 4–8 cases each, too few for conclusions.

In women aged 65 or older (45 cases), the full model's AUROC was 0.836 against 0.811 for OST (difference 0.025, −0.019 to 0.068) (Table 5); OST < 2 referred 71.1% of these women to detect 42 cases, and the full model at its global threshold referred 55.0% to detect 43. In women aged 50–64 (27 cases), SCORE had the highest AUROC (0.838 vs 0.818). In men aged 50–69 the full model exceeded OST by 0.080 (0.013–0.152), but from only 9 cases; in men aged 70 or older (15 cases) the full model, the three-variable model, OST and SCORE discriminated similarly (AUROC 0.732–0.756) and ORAI worse (0.629), but OST < 2 detected 14 cases by referring 59.8% of these men, whereas the full model at its global threshold detected 10 while referring 30.3%. With all five cycles (Supplementary Table S10), the full model exceeded OST by 0.031 (0.009–0.054) in women aged 50–64 (95 cases) and 0.029 (0.009–0.048) in women aged 65 or

older (223 cases); in men the differences were 0.025 (−0.024 to 0.078; 45 cases) at 50–69 years and 0.019 (−0.015 to 0.056; 76 cases) at 70 years or older. At 20% referral within each stratum, differences in cases detected were small and their CIs included zero. All stratum comparisons are exploratory.

Table 5: Performance in clinical strata of the temporal validation set. AUROC of each model; Δ, full model minus OST (paired bootstrap 95% CI). The last two columns give cases detected and the proportion referred by the full model at its global Youden threshold (4.8%) and by OST < 2 (no fitting). Exploratory; not part of the testing hierarchy.

			AUROC						Detected, referred	
Stratum	*n*	Cases	Full	Age, weight, sex	OST	SCORE	ORAI	Δ vs OST (95% CI)	Full model	OST < 2
Women 50–64 years	547	27	0.818	0.804	0.804	0.838	0.749	0.014 (−0.030 to 0.065)	18/27, 27.4%	21/27, 35.5%
Women ≥65 years	311	45	0.836	0.811	0.811	0.807	0.783	0.025 (−0.019 to 0.068)	43/45, 55.0%	42/45, 71.1%
Men 50–69 years	787	9	0.813	0.733	0.734	0.751	0.787	0.080 (0.013–0.152)	6/9, 5.8%	6/9, 19.4%
Men ≥70 years	363	15	0.741	0.732	0.733	0.756	0.629	0.008 (−0.074 to 0.100)	10/15, 30.3%	14/15, 59.8%

3.8 Sensitivity analyses

The conclusions did not change in any sensitivity analysis (Supplementary Tables S21 and S22). For the spine-inclusive outcomes the full model achieved AUROCs of 0.846 and 0.840, against 0.823 and 0.801 for OST; with diagnosed or treated participants included, 0.859 against 0.824. Shifting the pre-2013 T-scores down by 0.15 SD raised development prevalence to 5.5% and gave AUROCs of 0.871 (full model), 0.851 (three-variable model) and 0.841 (OST); shifting the 2013–2018 T-scores up instead gave 0.885, 0.859 and 0.849, with calibration slopes of 1.00–1.08 in both scenarios. Refitting without non-Hispanic Asian participants (0.861), on 2013–2014 alone (0.854), without 2005–2006 (0.869), without laboratory predictors (0.863) or without alcohol (0.869) made little difference. Across all versions of the analysis that included OST, the gain of the best model over OST was 0.028–0.045 (Supplementary Table S3).

4 Discussion

4.1 Principal findings

In 9,657 older US adults neither diagnosed with nor treated for osteoporosis, a penalised logistic regression using 35 history, examination, laboratory and socio-demographic predictors identified hip osteoporosis with a pooled AUROC of 0.881 across five held-out survey cycles, with tuning kept inside each development set (0.869 in the latest cycle, with good calibration). It exceeded OST by 0.024 and an age–weight–sex model by 0.019 in AUROC: consistently, but by an amount whose clinical relevance is uncertain, since the CIs span our margin of 0.02. When every single-learner family was tuned with the same budget, five of six ML families were equivalent to logistic regression, random forest was not shown to be equivalent, and none was better. At a fixed DXA capacity, the gain over OST was about 3 additional cases per 1,000 people screened (7%–9% more cases); about 60% of it was obtained by a model of clinical items alone, without blood tests, race or income.

4.2 Comparison with previous studies

Our results agree with systematic reviews concluding that simple tools based on age and weight perform as well as more complex instruments [11, 12]. OST's pooled AUROC across held-out cycles of 0.856 (0.841 in 2017–2018, and 0.834 in women in that cycle) lies at the upper end of the range reported in women [5, 9, 10] and is consistent with its performance in men [13, 14, 15]. The relative performance of the scores also matched previous comparisons: SCORE performed like OST, and ORAI, which groups age and weight into coarse bands, performed worse [10].

The ML studies on KNHANES found that ML did not clearly outperform logistic regression [18], as we did. NHANES-based models have reported AUROCs of 0.81–0.85 [20, 21, 22], similar to ours, but with random splits and, in most, without a simple comparator; where OST was included, the reported gain of a gradient-boosting model was larger than ours (0.83 vs 0.78) [21]. Differences in outcome definition, in whether diagnosed or treated people were excluded, and in how carefully each model was tuned may explain this. In our re-implementation of the feature-selection approach of Zhang et al. [22], all 12 combinations had an accuracy of 0.954–0.957, close to the 95.5% obtained by predicting "no osteoporosis" for everyone, including SVMs with AUROC as low as 0.653 (Supplementary Table S23). Accuracy therefore cannot rank osteoporosis models when the condition is uncommon.

4.3 How much do additional predictors and machine learning add?

Most of the discriminative information lay in age, weight and sex; further predictor groups added little individually and 0.018 together (Supplementary Table S13). Notably, the FRAX-type clinical risk factors (prior fracture, parental hip fracture, glucocorticoids, smoking and alcohol) and medical history added only 0.001 beyond body size and socio-demographic factors, and a short form without blood tests performed exactly like age, weight and sex. These relations are largely monotonic and additive, so flexible models have little structure to exploit: tuned XGBoost and LightGBM chose stumps (depth 1, two leaves), CatBoost shallow trees (depth 4), and spline terms added nothing. Equal tuning was essential to this conclusion. With default settings and its best imbalance strategy, each ML family trailed LR by up to 0.035 in cross-validated AUROC (SVM by 0.091 without one), and tuning closed the gap (Supplementary Table S8); comparing a tuned ML model with an untuned regression, or the reverse, could mislead either way [23].

Our margin of ± 0.02 was a judgement, not an external standard, and it cuts both ways. By it, the ML families were equivalent to logistic regression, whereas the full model was better than OST and age, weight and sex but neither equivalent to them nor demonstrably better by a relevant amount. Failing an equivalence test is not evidence of a meaningful difference; the gain was consistent across cycles, but its size is uncertain. In clinical terms [45], that gain was about 3 additional cases per 1,000 people screened at the same DXA capacity, 7%–9% more cases, or one additional case for every 330–430 people screened. Whether this justifies a tool that needs 35 inputs, including a biochemistry panel, vitamin D and income, is a value judgement that depends on how much an additional case is worth relative to the cost of collecting the inputs, which we did not model. A model of 19 clinical items from history and examination (body measures, menopause, hormone use, fractures, smoking, alcohol, glucocorticoids and common

diagnoses), without blood tests, race or income, obtained about two-thirds of the AUROC gain and 60% of the gain in cases; where these items are reliably recorded, part of the gain may be had at little cost, although several of them (parental hip fracture, glucocorticoid use, alcohol) are often missing or inaccurate in routine records, and a model built from such records needs its own validation. Laboratory tests and race, income and education each added a further 0.005 in AUROC and up to about 1.2 cases per 1,000 (Table 4). A sex-specific OST, which costs nothing to implement, recovered about a quarter of the AUROC gain but few cases (−0.1 to 1.0 per 1,000). Why the SHAP-selected short form performed no better than age, weight and sex while the 19-item clinical model did is unclear; selection by mean SHAP value may have discarded weakly informative but useful items, which argues against such data-driven short forms.

4.4 Why not FRAX?

FRAX estimates the 10-year probability of fracture [59] and is the standard tool for treatment decisions, but the question addressed here is who should have a DXA scan to find osteoporosis. For that purpose, FRAX calculated without BMD has discriminated low BMD less well than OST or SCORE in women aged 50–64 years (AUROC 0.60 vs 0.72–0.73) [17] and in older men (0.62 vs 0.68) [15]. NHANES also lacks several FRAX inputs in the required form (for example, current glucocorticoid use and secondary osteoporosis). We therefore chose OST, ORAI and SCORE, which were designed to predict low BMD, as the comparators.

4.5 Calibration, thresholds and implementation

Class weighting and SMOTE inflated predicted risks, roughly quadrupling the Brier score of LR until recalibration removed the difference, consistent with van den Goorbergh et al. [24]. When absolute risks are needed, models should be fitted on the original class distribution, with a threshold chosen for the intended use. The choice of threshold mattered more than the choice of model: a threshold set for 90% sensitivity in development detected 86 of 96 cases while referring 39% of participants (Supplementary Table S24). The operating point should be set locally from referral capacity and the relative costs of missed and unnecessary scans, and the cases-per-1,000 comparison and decision curves give the information to do so. Calibration also varied between held-out cycles (observed-to-expected ratio 0.84–1.16, slope 0.82–1.25), so the intercept should be recalibrated locally before predicted risks are used as absolute risks.

Age, sex, weight and height are recorded at almost every primary-care visit, so OST, a sex-specific OST and the three-variable model can be computed automatically in electronic health records. The full model also needs blood tests that are available only for patients who happen to have been tested, and income, which is rarely recorded. Any of these models can be implemented as a single linear equation (Supplementary Table S25); gradient-boosting models cannot, and would need software integration for no gain. We also derived a 10-predictor short form without blood tests, but it performed no better than age, weight and sex, and its face validity is limited: it omits sex and relies on postmenopausal status, which is coded 0 for all men, so in men it reduces to a model of body size, age, prior fracture and a few diagnoses (Supplementary Results). We therefore do not recommend it; where a simple tool is wanted, OST, a sex-specific OST or the three-variable model is preferable.

4.6 Subgroups and fairness

The model ranked men about as well as women, but a single threshold detected only two-thirds of cases in men, reflecting their low baseline risk; sex-specific thresholds improved this without increasing total referrals (Supplementary Results). Raising detection in men further would require more referrals, a policy decision about men, who are already under-screened [4]. In women aged 65 or older, for whom DXA is recommended regardless of risk scores, a tool is useful mainly for ordering patients when capacity is limited. In the strata where a tool would decide who is scanned at all, younger postmenopausal women and men, the full model's AUROC gain over OST across all five cycles (45–95 cases per stratum) was similar to that in older women, but the differences in cases detected at a fixed referral rate were small and their CIs included zero, so these data cannot choose between tools there. Removing race/ethnicity changed discrimination little and shifted predicted risk towards the pooled average, raising it in non-Hispanic Black adults and lowering it in white adults; but the model with race was itself miscalibrated in small groups (for example, 2.1% predicted vs 1.4% observed in non-Hispanic Black adults), and with 4–8 cases per group no trade-off between race-free prediction and calibration can be demonstrated [60, 40]. With only 4–8 cases in most minority groups, our data cannot show whether any tool detects osteoporosis less well in a particular group.

4.7 Strengths and limitations

Strengths include a large national probability sample with standardised DXA, restricted to the intended screening population; IECV across all five cycles, with the tuning of every model family nested inside each development set and equivalence tests applied in both directions, complemented by a temporal split and a bootstrap including model refitting; head-to-head comparison with three established screening indices and a sex-specific OST; assessment of calibration, net benefit and cases detected at fixed capacity; and full reproducibility from public data and open code, with every analysis version reported.

The study also has limitations. First, it is cross-sectional: the models detect existing osteoporosis and do not predict fractures. Second, the outcome and cohort were revised, and the testing hierarchy, benchmarks and IECV added, after the validation data had been examined in earlier versions; all versions are reported (Supplementary Table S3), but the hierarchy was not prespecified and the temporal validation is not strictly untouched, which is why we give more weight to IECV. Third, the hip-only outcome underestimates osteoporosis as defined clinically from the lowest T-score at any site [2]; spine-inclusive outcomes gave similar conclusions. Fourth, DXA scanners, laboratory methods and the alcohol questionnaire changed between cycles; analyses shifting T-scores by the observed 0.15 SD, without laboratory predictors or alcohol, and with development on the same scanner type gave similar results. Fifth, risk factors and prior diagnosis were self-reported, prior fractures were of any cause and glucocorticoid use was ever rather than current use, and ORAI and SCORE had to be adapted to the available items, which may understate their performance. Sixth, models were fitted without survey weights, and the non-response adjustment used few characteristics; participants without DXA were heavier and more often women and non-Hispanic Black, so the cohort may not fully represent the US population aged 50 or older. Seventh, missing values were imputed once, which understates

variance slightly; with 96 validation events, differences of a few hundredths in AUROC cannot be confirmed or excluded from the temporal split alone; the IECV development sets overlap, so its pooled CIs are approximate even with the HKSJ adjustment, and its case-count CIs are conditional on the fitted models; and men, younger women and minority groups had few cases. Eighth, every analysis reused data that had already been examined: no untouched validation data exist, because the 2021–2023 NHANES cycle has released no femur DXA data at the time of writing; the frozen model and code are public so that it can be validated when such data become available. Finally, all data are from the United States; external validation is needed, particularly where DXA access is most limited.

5 Conclusion

In older US adults neither diagnosed with nor treated for osteoporosis, equally tuned machine learning did not outperform logistic regression, and 35 clinical, laboratory and socio-demographic predictors added about 0.02 in AUROC to OST and to age, weight and sex, or about 3 more cases detected per 1,000 people screened at the same DXA capacity. The gain is consistent but small and of uncertain clinical relevance, and a model of clinical items without blood tests, race or income obtained about two-thirds of the AUROC gain over OST and 60% of the gain in cases. For osteoporosis case-finding, simple tools capture most of the achievable discrimination, and whether the remaining gain is worth a complex tool is a decision for each health system. The choice of threshold, and whether it is applied to men, matters more than the algorithm, and new models should be judged against these simple tools in the intended population.

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Supplementary material

Supplementary Methods

Exclusions. Osteoporosis medications were bisphosphonates (alendronate, risedronate, ibandronate, zoledronic acid, etidronate), denosumab, raloxifene, bazedoxifene, teriparatide, abaloparatide, romosozumab and calcitonin; 486 excluded participants used one. Participants with a missing answer to the diagnosis question were treated as not diagnosed.

Outcome. T-scores were (BMD − reference mean)/reference SD, with reference means of 0.858 g/cm^2 (SD 0.120) at the femoral neck and 0.942 g/cm^2 (SD 0.122) at the total hip [28]. Mean femoral neck BMD in women was 0.744–0.753 g/cm^2 in 2005–2010 and 0.729–0.731 g/cm^2 in 2013–2018. The spine-inclusive outcomes were (i) hip or complete-scan L1–L4 T-score ≤ -2.5 and (ii) hip or spine T-score ≤ -2.5 with spine BMD from at least two valid vertebrae (total bone mineral content divided by total projected area) [30], using the Hologic young white-female spine reference (mean 1.047 g/cm^2, SD 0.110) [2, 31].

Harmonisation. Age is top-coded at 85 years in 2005–2006 and at 80 years thereafter, so it was capped at 80 in all cycles. Creatinine from 2005–2006 was recalibrated to the standardised method ($-0.016 + 0.978\times$ measured value). Vitamin D was the LC-MS/MS or LC-MS/MS-equivalent value released for each cycle. Average daily alcohol intake was drinks per drinking day multiplied by drinking days per year, divided by 365; drinking frequency was recorded as a count up to 2013–2014 and as a category in 2017–2018, which we converted to approximate days per year. Values stored as numerical underflow ($< 10^{-30}$) were restored to 0. Other laboratory measurements were used as released; their means differed between periods (for example, alkaline phosphatase 72.6 vs 82.6 U/L; Supplementary Table S6), so we also refitted the model without laboratory predictors. Postmenopausal status was defined as no regular menstrual periods in the past 12 months (which also captures hysterectomy). Responses of "refused" or "don't know" were set to missing, and borderline diabetes was coded as no diabetes. The analysis stopped automatically if any predictor was entirely missing within a cycle. All predictors were standardised after imputation.

Benchmark scores. ORAI [7]: 15, 9 or 5 points for age ≥75, 65–74 or 55–64 years; 9 or 3 points for weight <60 or 60–69 kg; and 2 points for no estrogen use (NHANES records ever rather than current use of female hormones). SCORE [8]: 5 points if not Black, 4 for rheumatoid arthritis (from the arthritis-type question), 4 for each fracture site (maximum 12; NHANES records hip, wrist and spine fractures of any cause, rather than non-traumatic wrist, rib and hip fractures after age 45), 3 times the first digit of age and 1 for no estrogen use, minus weight in pounds divided by 10 (truncated). Rheumatoid arthritis was used only for SCORE. For the Boruta + XGBoost pipeline, Boruta [39] selected features from our candidate set on the development data, and XGBoost, tuned with the same budget as the other families, was fitted to them; because the original feature set and outcome differ, this tests the approach rather than the published model.

Re-implementation of the published analysis. To mirror Zhang et al. [22], we evaluated all combinations of three feature-selection methods (Boruta, Lasso [61] and XGBoost importance) with four classifiers (LR, SVM, RF and XGBoost) by 10-fold cross-validation on the development set. Features were selected once on the full development set, so these estimates are optimistic.

Model details. LR-spline used four knots at quantiles with linear extrapolation beyond the boundary knots. The stacking ensemble used RF, XGBoost, LightGBM and CatBoost as base learners with an LR meta-learner. SMOTE-Tomek [62], used in an earlier version, removed no Tomek links and was identical to SMOTE. Tuned hyperparameters were kept only if they improved on the defaults. Internal calibration was assessed with cross-fitted recalibration. Tuning and selection on the same folds make the internal estimates slightly optimistic but do not affect the temporal estimates.

Incremental value of predictor groups. Nested LR models added predictor groups in clinical order (age, weight and sex; other body measures and socio-demographic factors; clinical risk factors and history; laboratory tests; race/ethnicity), each developed and validated as the full model.

Predictor importance and short form. Importance was ranked by mean absolute SHAP value [57] in 500 randomly selected development-set participants (permutation explainer, 100 background samples, probability scale before recalibration). A short form was restricted to predictors a clinician can obtain at a visit without a blood test and that are not socially assigned (race/ethnicity, PIR and education excluded, and BMI as redundant with weight and height); the 10 most important remaining predictors were fitted with LR and evaluated as the full model.

Supplementary Results

Re-implementation: results. In 10-fold cross-validation, LR had the highest AUROC with every feature set (0.871–0.879), followed by RF and XGBoost (0.845–0.855, with no consistent order); SVM with default settings performed worst (0.653–0.777). Accuracy was 0.954–0.957 for all 12 combinations, almost identical to the 95.5% of the development set without osteoporosis (Supplementary Tables S26 and S23).

Incremental value: results. Age, weight and sex alone gave a temporal-validation AUROC of 0.851; no further predictor group produced a change whose CI excluded zero, and all 32 additional predictors together added 0.018 (Supplementary Table S13).

Sex-specific thresholds. Sex-specific Youden thresholds were 6.7% for women and 3.5% for men. For men, sensitivity rose from 0.667 to 0.750 (18 of 24 cases) and referrals from 13.6% to 17.9%; for women, referrals fell from 37.4% to 29.4% and sensitivity from 0.847 to 0.819 (59 of 72). Overall, the same 77 cases were detected with fewer referrals (458 vs 477). Sensitivity rose with age, from 0.656 at 50–64 years to 0.931 at 80 years or older (Supplementary

Table S20). Removing race shifted predicted risk towards the pooled average: risk was over-predicted in non-Hispanic Black adults (3.2% predicted vs 1.4% observed, against 2.1% with race) and under-predicted in non-Hispanic white adults (5.5% vs 7.1%, against 6.3%).

OST by sex. In men, OST < -1 detected 12 of 24 cases (sensitivity 0.500) while referring 11.2%, and OST < 2 detected 20 of 24 while referring 32.2% (Supplementary Table S27).

Predictor importance and short form: results. The most important predictors by mean absolute SHAP value were the body-size measures, postmenopausal status and age; no laboratory test ranked in the top ten (Supplementary Table S28, Supplementary Figures S2, S3 and S4). The 10-predictor short form (weight, postmenopausal status, age, height, waist, hormone use, prior fracture, arthritis, cancer and stroke) had a validation AUROC of 0.852 (0.810–0.887; slope 0.96), 0.018 (0.003–0.033) lower than the full model and the same as the three-variable model (difference 0.000, −0.013 to 0.014) (Supplementary Tables S29 and S30). Its sensitivity was 0.889 in women but 0.583 in men (14 of 24). It contains no sex term, and postmenopausal status and hormone use are 0 for all men, so in men it reduces to a function of body size, age, prior fracture and three diagnoses; the negative coefficients for arthritis and cancer reflect associations conditional on body size and age and should not be interpreted causally. We do not recommend it for use.

Table S1: TRIPOD+AI checklist [25]: where each item is reported.

Item	Topic	Where reported
1	Title	Title page
2	Abstract	Abstract
3a	Healthcare context and rationale	Introduction, paragraphs 1–3
3b	Intended purpose and users	Introduction; Discussion, Calibration, thresholds and implementation
3c	Known health inequalities	Introduction, paragraph 3; Methods, Subgroups, clinical strata and sensitivity analyses
4	Objectives	Introduction, final paragraph
5a	Data sources	Methods, Data source and study population
5b	Dates of data collection	Methods, Data source and study population; Development and validation sets
6a	Setting	Methods, Data source and study population
6b	Eligibility criteria	Methods, Data source and study population; Figure 1
6c	Treatments received	Methods, Data source and study population (diagnosed or treated excluded)
7	Data preparation and quality checks	Methods, Predictors and data preparation; Supplementary Methods; Table S4
8a	Outcome definition and assessment	Methods, Outcome
8b	Outcome assessors	Methods, Outcome (NHANES DXA technologists)
8c	Blinding of outcome assessment	Methods, Outcome (DXA independent of predictors)

Item	Topic	Where reported
9a	Choice of predictors	Methods, Predictors and data preparation
9b	Predictor definitions and measurement	Methods, Predictors and data preparation; Supplementary Methods
9c	Predictor assessors and blinding	Standard NHANES interview, examination and laboratory protocols
10	Sample size	Methods, Development and validation sets
11	Missing data	Methods, Missing data; Table S4
12a	Use of data in development and evaluation	Methods, Development and validation sets
12b	Handling of predictors	Methods, Missing data; Models and tuning; Supplementary Methods
12c	Model type, building and tuning	Methods, Models and tuning; Supplementary Tables S8 and S5
12d	Heterogeneity across clusters	Methods, Internal–external validation; Survey weighting and non-response; Subgroups, clinical strata and sensitivity analyses
12e	Performance measures	Methods, Recalibration, thresholds and performance measures
12f	Model updating or recalibration	Methods, Recalibration, thresholds and performance measures
12g	How predictions are calculated	Supplementary Tables S25 and S29; code
13	Class imbalance	Methods, Models and tuning
14	Fairness	Methods, Subgroups, clinical strata and sensitivity analyses
15	Model output and thresholds	Methods, Recalibration, thresholds and performance measures; Supplementary Table S24
16	Differences between development and evaluation data	Results, Participants; Supplementary Table S6
17	Ethics	Declarations
18a–b	Funding; conflicts of interest	Declarations
18c–d	Protocol; registration	Declarations (none)
18e–f	Data and code availability	Declarations
19	Patient and public involvement	Declarations (none)
20a	Participant flow	Figure 1
20b	Participant characteristics	Table 1
20c	Comparison of development and evaluation data	Supplementary Table S6
21	Numbers of participants and outcome events	Figure 1; Methods, Development and validation sets; Table S20
22	Full model specification	Supplementary Table S25; code repository
23a	Performance with confidence intervals, including subgroups	Tables 3, 4 and 5; Figures 2–4; Supplementary Table S20
23b	Heterogeneity across clusters	Table 2; Supplementary Tables S21, S22, S18, S19 and S12
24	Model updating results	Results, Calibration and clinical utility
25	Interpretation	Discussion

Item	Topic	Where reported
26	Limitations	Discussion, Strengths and limitations; Supplementary Table S3
27a	Handling of poor-quality or missing input data	Methods, Missing data (imputation model in code)
27b	User interaction and required expertise	Discussion, Calibration, thresholds and implementation
27c	Future research and applicability	Discussion, Calibration, thresholds and implementation; Strengths and limitations

Table S2: Participants aged ≥50 years in the cohort versus those excluded for lack of valid femoral neck DXA. Means or proportions; P from Welch's t-test for continuous variables and χ^2 tests for proportions.

Characteristic	Cohort (n = 9,657)	No DXA (n = 3,185)	Missing in no-DXA group	P
Age, years	64.2	67.6	0.0%	<0.001
Female sex	43.0%	59.1%	0.0%	<0.001
BMI, kg/m^2	28.7	33.0	25.1%	<0.001
Weight, kg	80.6	88.7	24.0%	<0.001
Prior hip, wrist or spine fracture	12.1%	15.6%	0.1%	<0.001
Ever daily glucocorticoid use	5.4%	7.7%	1.4%	<0.001
Non-Hispanic Black	21.1%	24.8%	0.0%	<0.001

Table S3: Temporal-validation results in every version of this analysis. Versions differed in outcome definition, cohort and comparators, as listed; all used the same 2005–2014 / 2017–2018 split. The present analysis is version 10.

Version	Outcome	Cohort and predictors	Prevalence	Best model AUROC (95% CI)	OST	LR: age, weight, sex
1	Femoral neck or L1–L4 (complete scans)	All with DXA; 29 predictors	9.2%	LR 0.843 (0.813–0.871)	–	–
2	Femoral neck or spine (≥2 vertebrae)	All with DXA; 35 predictors	10.8%	LR 0.839 (0.813–0.864)	0.796	0.822
3	Hip (femoral neck or total hip)	All with DXA; 35 predictors	6.6%	LR-spline 0.869 (0.839–0.895)	0.824	0.839
4	Hip	Previously diagnosed excluded; 35 predictors	4.7%	LR 0.870 (0.835–0.905)	0.839	0.850
5	Hip	Diagnosed or treated excluded; all families tuned	4.5%	LR 0.869 (0.831–0.903)	0.841	0.851
6	Hip	As 5; 40 trials for every family, wider search spaces; ORAI, SCORE; IECV	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851
7	Hip	As 6; secondary family of four; sex-specific OST; cases per 1,000; weight trimming; scanner shift	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851
8	Hip	As 7; IECV with nested tuning and HKSJ CIs; race/SES-free model; cases per 1,000 from IECV; 1,000-replicate refit bootstrap	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851
9	Hip	As 8; primary estimand IECV; clinical-only model; strata, weighted and random-tie case counts	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851
10	Hip	As 9; ML families re-tuned inside each IECV development set	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851

Table S4: Percentage of missing values by cycle (final cohort). Predictors not shown had ≤1% missing in every cycle.

Predictor	2005–06	2007–08	2009–10	2013–14	2017–18
Income-to-poverty ratio	4.9	9.8	9.9	8.2	13.4
25(OH) vitamin D, nmol/L	2.6	14.6	4.1	3.2	3.5
Biochemistry panel (range)	3.0–3.2	4.5–4.6	4.7–4.8	4.0–4.3	4.7–4.8
Waist, cm	1.1	1.1	1.1	1.6	1.2
Parental hip fracture	2.6	2.7	2.7	2.4	1.5
Postmenopausal	1.6	1.7	3.0	1.4	2.7
Ever used female hormones	1.8	1.9	3.3	1.6	2.9
Alcohol ≥3 drinks/day	2.9	3.5	5.7	3.9	4.7

Table S5: Hyperparameter search spaces and selected values (Optuna, tree-structured Parzen estimator, 40 trials per family, 5-fold CV AUROC in the development set). *Selected value within 5% of the search range from a bound (4 of the tuned hyperparameters). For XGBoost depth 1 and LightGBM two leaves, the bound is the smallest possible tree (a stump); LightGBM's minimum child size of 198 lies at the upper bound and indicates strong regularisation.

Family	Hyperparameter	Search range (scale)	Selected
LR	C	0.001–100 (log)	0.135
LR-spline	C	0.001–100 (log)	1.92
SVM	C	0.01–100 (log)	1.05
	gamma	1.00e-05–1 (log)	0.00315
RF	n estimators	200–1000 (integer, step 100)	700
	max depth	3–20 (integer, step 1)	12
	min samples leaf	1–60 (integer, step 1)	27
	max features	0.05–1 (linear)	0.0934*
XGBoost	n estimators	100–2000 (integer, step 50)	1600
	learning rate	0.002–0.3 (log)	0.0234
	max depth	1–8 (integer, step 1)	1*
	subsample	0.5–1 (linear)	0.706
	colsample bytree	0.5–1 (linear)	0.648
	min child weight	1–20 (log)	1.4
	reg lambda	0.001–10 (log)	2.9
LightGBM	n estimators	100–2000 (integer, step 50)	1300
	learning rate	0.002–0.3 (log)	0.0165
	num leaves	2–64 (integer, step 1)	2*
	min child samples	1–200 (integer, step 1)	198*
	subsample	0.5–1 (linear)	0.833
	colsample bytree	0.5–1 (linear)	0.614
	reg lambda	0.001–10 (log)	0.168
CatBoost	iterations	200–2000 (integer, step 100)	600
	depth	1–8 (integer, step 1)	4
	learning rate	0.002–0.3 (log)	0.0135
	l2 leaf reg	1–10 (log)	5.77

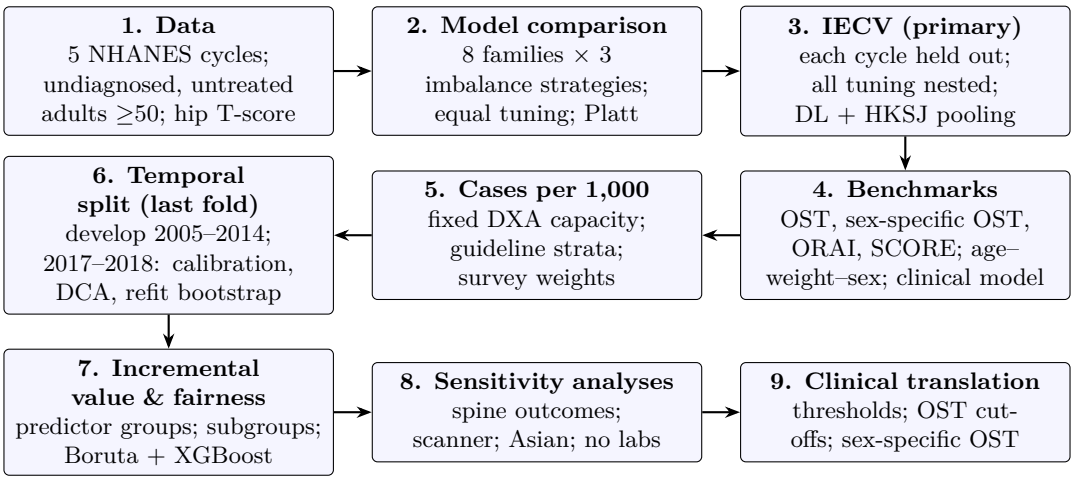


Figure S1: Study workflow. Imputation, scaling and resampling were fitted inside cross-validation folds. In the temporal split, the 2017–2018 cycle was used only for evaluation; in IECV each cycle was held out in turn.

Table S6: Characteristics of the development (2005–2014) and temporal validation (2017–2018) sets. Mean (SD) or n (%); unweighted tests.

Characteristic	Development (n = 7,649)	Validation (n = 2,008)	P
Osteoporosis (hip)	343 (4.5)	96 (4.8)	
Age, years	64.25 (9.48)	64.07 (8.91)	0.43
Income-to-poverty ratio	2.72 (1.62)	2.69 (1.60)	0.48
Education level (1–5)	3.22 (1.34)	3.48 (1.25)	<0.001
BMI, kg/m^2	28.63 (5.46)	29.14 (5.94)	<0.001
Weight, kg	80.51 (17.70)	81.11 (19.22)	0.21
Height, cm	167.50 (9.97)	166.53 (9.81)	<0.001
Waist, cm	100.95 (13.62)	101.77 (14.39)	0.022
Calcium, mg/dL	9.46 (0.38)	9.32 (0.37)	<0.001
Phosphorus, mg/dL	3.74 (0.56)	3.54 (0.53)	<0.001
Alkaline phosphatase, U/L	72.64 (24.37)	82.57 (25.16)	<0.001
Creatinine, mg/dL	0.99 (0.54)	0.96 (0.41)	0.031
Albumin, g/dL	4.20 (0.30)	4.04 (0.30)	<0.001
Uric acid, mg/dL	5.71 (1.42)	5.66 (1.47)	0.21
Glucose, mg/dL	110.18 (44.03)	109.68 (43.61)	0.66
Total protein, g/dL	7.12 (0.50)	7.12 (0.44)	0.67
Cholesterol, mg/dL	199.23 (43.63)	192.77 (44.14)	<0.001
25(OH) vitamin D, nmol/L	65.06 (25.99)	75.84 (32.93)	<0.001
Female sex	3,293 (43.1)	858 (42.7)	0.81
Diabetes	1,380 (18.1)	468 (23.3)	<0.001
Arthritis	3,027 (39.7)	816 (40.7)	0.41
Stroke	429 (5.6)	128 (6.4)	0.22
Coronary heart disease	570 (7.5)	156 (7.8)	0.67
Cancer	1,181 (15.5)	305 (15.2)	0.80
Former smoker	2,722 (35.6)	621 (30.9)	<0.001
Current smoker	1,338 (17.5)	316 (15.7)	0.067
Prior hip, wrist or spine fracture	917 (12.0)	250 (12.5)	0.59
Parental hip fracture	716 (9.6)	218 (11.0)	0.068
Ever daily glucocorticoid use	390 (5.1)	127 (6.4)	0.032
Postmenopausal	2,889 (38.5)	755 (38.6)	0.95
Ever used female hormones	1,161 (15.5)	208 (10.7)	<0.001
Alcohol $\geq$3 drinks/day	168 (2.3)	57 (3.0)	0.097
Mexican American	1,118 (14.6)	231 (11.5)	<0.001
Other Hispanic	608 (7.9)	217 (10.8)	<0.001
Non-Hispanic Black	1,549 (20.3)	485 (24.2)	<0.001
Other/multiracial (incl. Asian)	453 (5.9)	357 (17.8)	<0.001

Table S7: Cross-validated performance of 24 model × imbalance-strategy configurations with default hyperparameters (all 35 predictors, 5-fold CV, development set). Brier scores are shown before (raw) and after cross-fitted Platt recalibration (recal.). The selected configuration is in bold.

				Brier						Brier	
Model	Strategy	AUROC	AUPRC	raw	recal.	Model	Strategy	AUROC	AUPRC	raw	recal.
LR	**None**	**0.876**	**0.346**	**0.035**	**0.035**	XGBoost	None	0.857	0.275	0.038	0.037
LR	Class wt	0.872	0.346	0.137	0.035	XGBoost	Class wt	0.848	0.262	0.058	0.037
LR	SMOTE	0.871	0.336	0.127	0.035	XGBoost	SMOTE	0.855	0.266	0.041	0.037
LR-spline	None	0.873	0.339	0.035	0.035	LightGBM	None	0.852	0.275	0.039	0.037
LR-spline	Class wt	0.870	0.337	0.133	0.035	LightGBM	Class wt	0.842	0.258	0.050	0.038
LR-spline	SMOTE	0.868	0.331	0.125	0.036	LightGBM	SMOTE	0.848	0.267	0.040	0.037
SVM	None	0.785	0.239	0.038	0.038	CatBoost	None	0.865	0.294	0.037	0.037
SVM	Class wt	0.841	0.238	0.038	0.038	CatBoost	Class wt	0.851	0.299	0.072	0.037
SVM	SMOTE	0.812	0.212	0.055	0.039	CatBoost	SMOTE	0.841	0.246	0.042	0.038
RF	None	0.853	0.298	0.037	0.037	Stacking	None	0.863	0.302	0.037	0.038
RF	Class wt	0.861	0.303	0.043	0.036	Stacking	Class wt	0.865	0.311	0.036	0.036
RF	SMOTE	0.861	0.250	0.058	0.038	Stacking	SMOTE	0.864	0.270	0.045	0.038

Table S8: Cross-validated AUROC of each model family before and after equal-budget tuning (Optuna, 40 trials per family, development set, same 5 folds), with its best imbalance strategy. Stacking used the tuned base learners. Changes are computed from unrounded values. The tuning estimate for LR (0.879) is higher than its cross-fitted internal estimate (0.877) because it was obtained on the folds used for tuning. Search spaces and selected values are in Supplementary Table S5.

Family	Strategy	Trials	Default AUROC	Tuned AUROC	Change
LR	None	40	0.876	0.879	0.002
SVM	Class wt	40	0.841	0.876	0.035
LR-spline	None	40	0.873	0.875	0.001
XGBoost	None	40	0.857	0.875	0.018
LightGBM	None	40	0.852	0.875	0.023
Stacking	Class wt	–	0.865	0.874	0.010
CatBoost	None	40	0.865	0.873	0.008
RF	Class wt	40	0.861	0.864	0.003

Table S9: Clinical model (19 predictors from history and examination; no laboratory tests, race/ethnicity, income or education) in internal–external cross-validation, with the penalty C re-tuned within the four development cycles. Δ, AUROC differences with 95% CIs (paired bootstrap per cycle; DerSimonian–Laird with HKSJ CIs when pooled). The pooled AUROC is meta-analysed across cycles; Supplementary Table S10 instead pools the out-of-cycle predictions (0.865).

Held-out cycle	C	AUROC	Slope	Δ vs OST	Δ vs age, weight, sex	Full model minus clinical
2005–2006	31.7	0.900	1.08	0.000 (−0.026 to 0.027)	−0.010 (−0.030 to 0.010)	0.012 (0.002 to 0.022)
2007–2008	0.26	0.897	1.26	0.030 (0.012 to 0.049)	0.025 (0.009 to 0.040)	0.003 (−0.011 to 0.018)
2009–2010	1.00	0.872	0.95	0.014 (−0.001 to 0.029)	0.019 (0.006 to 0.032)	0.005 (−0.008 to 0.018)
2013–2014	0.13	0.820	0.73	0.012 (−0.011 to 0.035)	0.008 (−0.011 to 0.028)	0.021 (0.008 to 0.035)
2017–2018	0.20	0.859	1.00	0.018 (−0.001 to 0.037)	0.008 (−0.008 to 0.024)	0.010 (−0.001 to 0.021)
Pooled (HKSJ)	–	0.871 (0.831–0.910)	–	0.016 (0.004 to 0.029)	0.011 (−0.005 to 0.027)	0.011 (0.003 to 0.019)

Table S10: Guideline strata, the population not routinely screened, and survey weighting, from the pooled out-of-cycle predictions of the internal–external cross-validation. AUROCs pool the out-of-cycle predictions of all cycles (one-stage), so they differ slightly from the meta-analysed estimates in Table 2 and Supplementary Table S9. AUROC difference, full model minus OST (bootstrap within cycles). Cases per 1,000 at a referral rate of 20% within each population (random tie-breaking); Δ, full model minus OST (95% CI). Weighted rows use the examination weights (divided by five when pooled); referral is the top 20% of the weighted population, and CIs are from a simple bootstrap within cycles, not a design-based bootstrap.

			AUROC				Cases per 1,000 at 20% referral			
Population	*n*	Cases	Full	Clinical	OST	ΔAUROC (95% CI)	Total	Full	OST	Δ (95% CI)
All participants	9,657	439	0.875	0.865	0.850	0.026 (0.015–0.036)	45.5	35.3	32.5	2.8 (1.0–4.3)
Not routinely screened (all but women ≥65)	7,954	216	0.848	0.833	0.819	0.029 (0.012–0.046)	27.2	20.7	18.5	2.3 (0.9–3.9)
Women 50–64	2,448	95	0.869	0.858	0.838	0.031 (0.009–0.054)	38.8	27.8	27.2	0.6 (−2.9 to 4.1)
Women ≥65	1,703	223	0.824	0.814	0.795	0.029 (0.009–0.048)	130.9	75.7	72.4	3.4 (−4.1 to 10.0)
Men 50–69	3,699	45	0.797	0.771	0.772	0.025 (−0.024 to 0.078)	12.2	7.8	8.1	−0.3 (−1.4 to 1.1)
Men ≥70	1,807	76	0.778	0.758	0.759	0.019 (−0.015 to 0.056)	42.1	27.1	24.9	2.2 (−2.8 to 6.6)
All cycles, survey-weighted	9,657	439					38.9	30.9	28.4	2.4 (0.5–4.5)
2017–2018, unweighted	2,008	96					47.8	36.9	34.9	2.0 (−1.5 to 5.0)
2017–2018, survey-weighted	2,008	96					41.6	34.6	30.5	4.1 (−0.7 to 9.8)

Table S11: Tuned ML families in internal–external cross-validation with fully nested tuning: for each held-out cycle, every family re-selected its imbalance strategy and was re-tuned (40 Optuna trials, same search spaces) on the other four cycles only, as was the penalty of the full model; stacking omitted. AUROC by held-out cycle and pooled (random effects); Δ, family minus full model, pooled with 90% HKSJ CI; equivalent if within ±0.02. The last column gives Δ when hyperparameters were instead fixed at their 2005–2014 values, which overlap four held-out cycles. Strategies selected: none for LR-spline, XGBoost and CatBoost; class weighting for SVM and RF; none for LightGBM except class weighting when 2013–2014 was held out.

		Held-out cycle					Nested tuning		Fixed hyperparameters
Family	Pooled	2005–06	2007–08	2009–10	2013–14	2017–18	Δ vs full model (90% CI)	Equivalent	Δ (90% CI)
Full model (LR, re-tuned)	0.881	0.912	0.901	0.877	0.841	0.869	–	–	–
LR-spline	0.879	0.908	0.892	0.884	0.839	0.869	−0.003 (−0.008 to 0.003)	Yes	0.000 (−0.006 to 0.006)
SVM	0.878	0.912	0.896	0.874	0.834	0.870	−0.003 (−0.007 to 0.001)	Yes	0.000 (−0.004 to 0.004)
RF	0.870	0.914	0.894	0.861	0.812	0.862	−0.010 (−0.021 to 0.000)	No	−0.013 (−0.025 to −0.001)
XGBoost	0.878	0.912	0.895	0.874	0.836	0.868	−0.003 (−0.010 to 0.003)	Yes	−0.003 (−0.009 to 0.004)
LightGBM	0.875	0.914	0.892	0.869	0.825	0.867	−0.006 (−0.012 to 0.001)	Yes	−0.003 (−0.009 to 0.004)
CatBoost	0.879	0.919	0.896	0.880	0.827	0.867	−0.002 (−0.009 to 0.005)	Yes	−0.002 (−0.010 to 0.005)

Table S12: Sex-specific OST and all tools by sex (temporal validation set). The sex-specific OST is a logistic model of OST, sex and their interaction fitted on the development set: logit risk = $-3.195 - 0.353\,\mathrm{OST} + 0.601\,\mathrm{female} - 0.070\,\mathrm{OST} \times \mathrm{female}$. Within each sex it ranks people as OST does, so its sex-specific AUROC equals that of OST. Δ, full model minus each tool (paired bootstrap 95% CI).

	Full model	Age, weight, sex	OST, sex-specific	OST	SCORE	ORAI
All participants (n = 2,008; 96 cases)						
AUROC (95% CI)	0.869 (0.831–0.904)	0.851 (0.808–0.890)	0.850 (0.806–0.890)	0.841 (0.796–0.883)	0.841 (0.799–0.879)	0.791 (0.742–0.834)
Tool minus sex-specific OST	0.019 (−0.0004 to 0.041)	0.001 (−0.002 to 0.005)	–	−0.009 (−0.017 to −0.002)	−0.010 (−0.034 to 0.016)	−0.059 (−0.090 to −0.032)
Detected / referred at Youden threshold	77 / 23.8%	75 / 24.4%	76 / 25.3%	79 / 30.9%	80 / 30.2%	72 / 28.8%
Women (n = 858; 72 cases)						
AUROC	0.853	0.834	0.834	0.834	0.843	0.802
Δ full model (95% CI)	–	0.019 (−0.005 to 0.047)		0.019 (−0.005 to 0.046)	0.011 (−0.020 to 0.042)	0.051 (0.021–0.086)
Men (n = 1,150; 24 cases)						
AUROC	0.820	0.800	0.800	0.800	0.806	0.779
Δ full model (95% CI)	–	0.021 (−0.030 to 0.069)		0.020 (−0.031 to 0.068)	0.014 (−0.038 to 0.068)	0.042 (−0.027 to 0.120)

Table S13: Incremental value of predictor groups in nested logistic models. CV, 5-fold cross-validation in the development set; temporal, 2017–2018 validation set. ΔAUROC compares each model with the previous row (paired bootstrap 95% CI and P).

Predictors	k	CV AUROC	Temporal AUROC (95% CI)	AUPRC	ΔAUROC (95% CI)	P
Age, weight, sex	3	0.856	0.851 (0.808–0.890)	0.312	–	–
+ other body measures, socio-demographics	8	0.864	0.858 (0.816–0.894)	0.332	0.007 (−0.005 to 0.018)	0.21
+ clinical risk factors and history	21	0.869	0.859 (0.819–0.894)	0.318	0.001 (−0.010 to 0.012)	0.85
+ laboratory tests	31	0.872	0.863 (0.825–0.898)	0.325	0.005 (−0.002 to 0.011)	0.17
+ race/ethnicity	35	0.877	0.869 (0.831–0.904)	0.335	0.006 (−0.001 to 0.012)	0.14

Table S14: Temporal validation performance of every model, including operating characteristics (n = 2,008; 96 with osteoporosis), with 95% bootstrap CIs. Sensitivity, specificity and the proportion referred are at each model's development-set Youden threshold; for OST, ORAI and SCORE, values other than AUROC and AUPRC are for the recalibrated score. All ML families were tuned with the same budget. ΔAUROC, full model minus each alternative (paired bootstrap; 95% percentile CI, except 90% CI for the equivalence family); P, two-sided bootstrap P; P_{Holm}, Holm-adjusted within its family. Equivalence margin ±0.02.

Model	AUROC (95% CI)	AUPRC	Brier	Sens.	Spec.	Referred	Slope (95% CI)	ΔAUROC (95% CI)	P	P_{Holm}
Full model (LR, 35 predictors)	**0.869 (0.831–0.904)**	**0.335**	**0.038**	**0.802**	**0.791**	**23.8%**	**1.00 (0.85–1.18)**	–	–	–
Primary comparison										
OST	0.841 (0.796–0.883)	0.280	0.040	0.823	0.717	30.9%	0.98 (0.79–1.20)	0.028 (0.007–0.052)	0.014	–
Secondary comparisons (Holm-adjusted, family of four)										
LR: age, weight, sex	0.851 (0.808–0.890)	0.312	0.039	0.781	0.783	24.4%	1.04 (0.86–1.28)	0.018 (−0.001 to 0.039)	0.070	0.14
SCORE	0.841 (0.799–0.879)	0.263	0.040	0.833	0.725	30.2%	1.04 (0.87–1.25)	0.029 (0.003–0.055)	0.028	0.084
ORAI	0.791 (0.742–0.834)	0.177	0.042	0.750	0.735	28.8%	0.93 (0.75–1.11)	0.078 (0.047–0.115)	0.001	0.004
Boruta + XGBoost pipeline	0.856 (0.820–0.892)	0.295	0.039	0.844	0.680	34.5%	1.04 (0.89–1.24)	0.013 (−0.007 to 0.030)	0.18	0.18
Tuned ML families vs LR: equivalence (90% CI, margin ±0.02; no multiplicity adjustment)										
LR-spline	0.869 (0.832–0.903)	0.320	0.038	0.802	0.793	23.6%	1.06 (0.91–1.25)	0.000 (−0.008 to 0.007)	0.94	–
SVM	0.870 (0.832–0.906)	0.367	0.037	0.740	0.835	19.2%	1.03 (0.87–1.22)	−0.001 (−0.010 to 0.009)	0.87	–
Random forest	0.862 (0.823–0.900)	0.313	0.038	0.812	0.742	28.4%	1.04 (0.87–1.24)	0.007 (−0.008 to 0.020)	0.40	–
XGBoost	0.868 (0.834–0.901)	0.320	0.038	0.771	0.828	20.0%	1.04 (0.90–1.22)	0.001 (−0.010 to 0.012)	0.88	–
LightGBM	0.867 (0.832–0.902)	0.325	0.038	0.802	0.745	28.1%	1.06 (0.92–1.26)	0.002 (−0.010 to 0.012)	0.83	–
CatBoost	0.867 (0.830–0.905)	0.333	0.038	0.823	0.741	28.6%	1.07 (0.92–1.26)	0.002 (−0.011 to 0.013)	0.86	–
Stacking	0.872 (0.836–0.908)	0.338	0.038	0.812	0.769	25.8%	1.04 (0.89–1.24)	−0.003 (−0.012 to 0.006)	0.51	–
Exploratory (not part of a comparison family)										
Full model without race	0.863 (0.825–0.898)	0.325	0.038	0.792	0.798	23.1%	1.00 (0.84–1.19)	0.006 (−0.001 to 0.012)	0.14	–
Short form (10 predictors, no blood test)	0.852 (0.810–0.887)	0.310	0.039	0.812	0.751	27.6%	0.96 (0.80–1.17)	0.018 (0.003–0.033)	0.018	–

Table S15: Cases detected per 1,000 people screened at fixed referral rates in the temporal validation set only (2017–2018; 96 cases, 47.8 per 1,000), with models developed on 2005–2014. Ties broken by participant order; breaking them at random (Supplementary Table S10) changed only the sex-specific OST at 150 scans (32.7; Δ 1.2). Δ, full model minus OST and minus sex-specific OST, with 95% bootstrap CIs.

	Cases detected per 1,000 screened						Δ full model (95% CI)	
DXA scans per 1,000	Full model	Age, weight, sex	OST, sex-specific	OST	SCORE	ORAI	vs OST	vs sex-specific OST
150	33.9	32.4	32.9	30.9	27.4	23.9	3.0 (−1.5 to 6.5)	1.0 (−3.5 to 5.0)
200	36.9	35.9	35.9	34.9	31.9	24.9	2.0 (−1.5 to 5.0)	1.0 (−2.5 to 4.0)
250	39.3	37.8	37.4	37.4	37.4	31.9	2.0 (−1.0 to 4.5)	2.0 (−1.0 to 4.0)
300	41.3	38.8	38.8	39.3	39.8	36.9	2.0 (−0.5 to 4.0)	2.5 (0.0 to 5.0)

Table S16: OST at its published cut-offs (no fitting) versus the full model at a threshold referring the same proportion of participants (temporal validation set, 96 cases). Additional cases: full model minus OST, with 95% bootstrap CI.

Referral rate	Rule	Cases detected	Sensitivity	Specificity	Additional cases
39.1%	OST < 2	83	0.865	0.633	
	Full model	86	0.896	0.634	3 (−2 to 8)
14.9%	OST < -1	62	0.646	0.876	
	Full model	68	0.708	0.879	6 (−2 to 14)

Table S17: Net benefit in the temporal validation set at selected threshold probabilities within the 2%–15% range (95% bootstrap CIs), with true and false positives per 100 people screened. OST, ORAI and SCORE recalibrated to probabilities. Every model had higher net benefit than referring everyone or no one at every threshold from 2% to 15%.

Model	Threshold	Net benefit (95% CI)	TP per 100	FP per 100
Full model	0.05	0.0279 (0.0194–0.0371)	3.78	18.97
	0.10	0.0215 (0.0148–0.0299)	3.14	8.86
	0.15	0.0122 (0.0059–0.0191)	2.19	5.48
Boruta + XGBoost	0.05	0.0259 (0.0176–0.0341)	3.64	19.92
	0.10	0.0190 (0.0122–0.0268)	2.89	8.91
	0.15	0.0146 (0.0071–0.0220)	2.44	5.53
LR: age, weight, sex	0.05	0.0265 (0.0181–0.0354)	3.69	19.62
	0.10	0.0181 (0.0106–0.0256)	2.84	9.26
	0.15	0.0114 (0.0056–0.0178)	1.89	4.28
OST (recalibrated)	0.05	0.0268 (0.0181–0.0356)	3.74	20.12
	0.10	0.0183 (0.0108–0.0261)	2.84	9.11
	0.15	0.0077 (0.0021–0.0140)	1.69	5.23
ORAI (recalibrated)	0.05	0.0202 (0.0128–0.0280)	3.09	20.27
	0.10	0.0122 (0.0064–0.0188)	1.99	6.97
	0.15	0.0048 (0.0011–0.0091)	0.80	1.79
SCORE (recalibrated)	0.05	0.0257 (0.0177–0.0345)	3.64	20.22
	0.10	0.0160 (0.0091–0.0231)	2.59	8.91
	0.15	0.0085 (0.0031–0.0144)	1.64	4.48

Table S18: Survey-weighted performance in the temporal validation set. Weighted: 2017–2018 examination weights, with design-based 95% CIs from a Rao–Wu rescaled bootstrap over primary sampling units within strata (500 replicates). Non-response adjusted: examination weights multiplied by the inverse of the modelled probability of having a valid DXA result. Δ, full model minus comparator (non-response-adjusted). Observed prevalence 4.2% with both weightings; mean predicted risk of the full model 4.1% (examination weights) and 3.9% (non-response adjusted).

Model	Weighted AUROC (95% CI)	Non-response-adjusted AUROC (95% CI)	Δ (95% CI)	Predicted prevalence (NR-adjusted)
Full model	0.889 (0.835–0.937)	0.872 (0.812–0.934)	–	3.9%
Full model without race	0.887 (0.833–0.933)	0.871 (0.802–0.933)	0.001 (−0.004 to 0.009)	3.7%
Boruta + XGBoost pipeline	0.872 (0.809–0.929)	0.860 (0.788–0.927)	0.012 (−0.014 to 0.040)	3.8%
LR: age, weight, sex	0.870 (0.822–0.917)	0.856 (0.793–0.912)	0.016 (−0.009 to 0.043)	3.5%
OST	0.868 (0.816–0.914)	0.849 (0.785–0.913)	0.023 (−0.003 to 0.051)	3.4%
ORAI	0.819 (0.762–0.874)	0.779 (0.688–0.874)	0.093 (0.043–0.147)	3.6%
SCORE	0.860 (0.807–0.904)	0.837 (0.762–0.906)	0.035 (0.004–0.070)	3.8%
LR-spline	0.881 (0.819–0.935)	0.866 (0.793–0.934)	0.006 (−0.007 to 0.019)	4.2%
SVM	0.885 (0.830–0.935)	0.873 (0.815–0.937)	−0.001 (−0.015 to 0.018)	4.0%
Random forest	0.877 (0.810–0.936)	0.870 (0.804–0.934)	0.002 (−0.028 to 0.029)	3.9%
XGBoost	0.883 (0.824–0.938)	0.871 (0.805–0.941)	0.001 (−0.017 to 0.020)	4.0%
LightGBM	0.885 (0.820–0.938)	0.874 (0.802–0.941)	−0.002 (−0.021 to 0.017)	3.9%
CatBoost	0.885 (0.830–0.940)	0.874 (0.812–0.935)	−0.002 (−0.024 to 0.018)	3.9%
Stacking	0.890 (0.836–0.937)	0.877 (0.812–0.940)	−0.005 (−0.018 to 0.006)	3.9%

Table S19: Sensitivity of survey-weighted estimates to the weights (temporal validation set). The non-response factor (inverse of the modelled probability of a valid DXA result) had a median of 1.17, a 99th percentile of 3.85 and a maximum of 32.5 (18 participants above 5, 11 above 10). Normalised non-response-adjusted weights ranged from 0.13 (1st percentile) to 6.67 (99th percentile), maximum 10.1; the Kish design effect due to weighting was 2.87 with examination weights and 2.84 with adjusted weights. Weighted AUROCs; point estimates. Δ is computed from unrounded AUROCs, so it may differ by 0.001 from the difference of the rounded values.

Weights	Observed prev.	Predicted prev. (full model)	Full model	Age, weight, sex	OST	Δ vs OST
Examination weight	4.2%	4.1%	0.889	0.870	0.868	0.020
Non-response adjusted	4.2%	3.9%	0.872	0.856	0.849	0.023
Adjusted, trimmed at 99th percentile	4.2%	4.0%	0.871	0.855	0.848	0.023
Adjusted, trimmed at 5 $\times$ median	4.7%	4.4%	0.870	0.851	0.845	0.025
Adjusted, factor capped at 3	4.1%	4.0%	0.888	0.869	0.867	0.021

Table S20: Performance of the full model by subgroup in the temporal validation set, at the global threshold (4.8%). Pred., mean predicted risk. 95% bootstrap CIs. The last three columns show the race-free model at its own threshold (5.2%). In the models, non-Hispanic Asian adults are part of the other/multiracial category (see Methods). [a]All 6 cases were detected, so every bootstrap resample gave a sensitivity of 1.000 and the percentile interval is degenerate; it does not indicate certainty.

				Full model				Race-free model		
Subgroup	n	Cases	Prev.	Pred.	AUROC (95% CI)	Detected, sensitivity (95% CI)	Spec.	Pred.	AUROC	Sens.
Women	858	72	8.4%	7.2%	0.853 (0.811–0.894)	61/72, 0.847 (0.754–0.922)	0.669	7.3%	0.847	0.847
Men	1,150	24	2.1%	2.4%	0.820 (0.733–0.899)	16/24, 0.667 (0.468–0.850)	0.876	2.5%	0.816	0.625
Age 50–64	1,157	32	2.8%	2.7%	0.861 (0.790–0.924)	21/32, 0.656 (0.500–0.818)	0.864	2.7%	0.849	0.625
Age 65–79	651	35	5.4%	5.0%	0.855 (0.787–0.912)	29/35, 0.829 (0.696–0.944)	0.752	5.2%	0.847	0.829
Age $\geq$80	200	29	14.5%	12.9%	0.793 (0.696–0.882)	27/29, 0.931 (0.833–1.000)	0.450	13.0%	0.800	0.931
Non-Hispanic white	718	51	7.1%	6.3%	0.851 (0.799–0.897)	42/51, 0.824 (0.717–0.917)	0.736	5.5%	0.855	0.804
Non-Hispanic Black	485	7	1.4%	2.1%	0.782 (0.570–0.930)	4/7, 0.571 (0.143–1.000)	0.874	3.2%	0.776	0.571
Mexican American	231	8	3.5%	2.7%	0.816 (0.674–0.956)	4/8, 0.500 (0.143–0.889)	0.883	3.7%	0.803	0.625
Other Hispanic	217	6	2.8%	3.8%	0.963 (0.930–0.987)	6/6, 1.000 (1.000–1.000)[a]	0.806	3.8%	0.957	1.000
Non-Hispanic Asian	275	20	7.3%	6.0%	0.904 (0.840–0.953)	18/20, 0.900 (0.750–1.000)	0.698	5.8%	0.904	0.850
Other/multiracial (non-Asian)	82	4	4.9%	3.9%	0.821 (0.450–0.988)	3/4, 0.750 (0.000–1.000)	0.744	3.8%	0.814	0.750

Table S21: Sensitivity analyses: temporal-validation AUROC and calibration slope.

Analysis / model	Dev. prev.	Val. prev.	AUROC	Slope
Hip or complete-scan lumbar spine (L1–L4)				
Full model	7.2%	6.7%	0.846	1.06
OST			0.823	1.11
LR: age, weight, sex			0.836	1.12
Hip or lumbar spine from ≥2 valid vertebrae				
Full model	8.4%	8.2%	0.840	1.05
OST			0.801	1.04
LR: age, weight, sex			0.824	1.10
Excluding non-Hispanic Asian participants				
Full model	4.4%	4.4%	0.861	0.92
Development on 2013–2014 only (same scanner type)				
Full model	4.8%	4.8%	0.854	1.13
Development set without 2005–2006				
Full model	4.5%	4.8%	0.869	1.00
Without laboratory predictors				
Full model	4.5%	4.8%	0.863	0.98
Without alcohol predictor				
Full model	4.5%	4.8%	0.869	0.99
Including previously diagnosed or treated participants				
Full model	6.4%	7.1%	0.859	0.97
OST			0.824	0.89
LR: age, weight, sex			0.839	0.95

Table S22: Sensitivity to a possible scanner offset. Hip T-scores of one scanner period were shifted by 0.15 reference SD (the observed difference in mean femoral neck BMD in women), the outcome was redefined and all models were refitted on the development set.

Scenario	Model	Dev. prev.	Val. prev.	AUROC	Slope
Pre-2013 T-scores lowered by 0.15 SD	Full model	5.5%	4.8%	0.871	1.01
	Age, weight, sex			0.851	1.08
	OST			0.841	1.02
2013–2018 T-scores raised by 0.15 SD	Full model	4.1%	3.5%	0.885	1.07
	Age, weight, sex			0.859	1.07
	OST			0.849	1.00

Table S23: Re-implementation of the published analysis: 10-fold cross-validation in the development set (mean ± SD across folds). Features were selected once on the full development set, so these estimates are optimistic.

Feature selection	Classifier	AUROC	Accuracy
Boruta	LR	0.871 ± 0.021	0.956
	SVM	0.657 ± 0.050	0.955
	RF	0.849 ± 0.020	0.956
	XGBoost	0.845 ± 0.019	0.954
Lasso	LR	0.879 ± 0.020	0.957
	SVM	0.777 ± 0.042	0.955
	RF	0.848 ± 0.020	0.956
	XGBoost	0.855 ± 0.016	0.956
XGBoost importance	LR	0.872 ± 0.017	0.956
	SVM	0.653 ± 0.086	0.956
	RF	0.854 ± 0.025	0.956
	XGBoost	0.852 ± 0.022	0.955

Table S24: Full model at alternative operating points in the temporal validation set (96 cases). The first two thresholds were chosen on the development set; the others are fixed predicted risks.

Operating point	Threshold	Referred	Detected	Sens.	Spec.	PPV	NPV
Youden (development)	4.8%	23.8%	77/96	0.802	0.791	0.161	0.988
Sensitivity 0.90 in development	2.3%	39.3%	86/96	0.896	0.632	0.109	0.992
Fixed 3%	3.0%	32.8%	83/96	0.865	0.699	0.126	0.990
Fixed 5%	5.0%	22.8%	76/96	0.792	0.801	0.166	0.987
Fixed 10%	10.0%	12.0%	63/96	0.656	0.907	0.261	0.981

Table S25: Coefficients of the full model (penalised logistic regression on 35 predictors, recalibrated, original units). Units as in Supplementary Table S6; binary predictors 1 = yes. Coefficients are penalised and correlated predictors are not individually interpretable.

Term	Coefficient	Odds ratio per unit
Intercept	6.95527	
Age, years	0.05908	1.061
Income-to-poverty ratio	−0.14411	0.866
Education level (1–5)	0.03807	1.039
BMI, kg/m^2	−0.15810	0.854
Weight, kg	−0.03937	0.961
Height, cm	−0.04479	0.956
Waist, cm	0.02589	1.026
Calcium, mg/dL	−0.01321	0.987
Phosphorus, mg/dL	−0.00500	0.995
Alkaline phosphatase, U/L	0.00771	1.008
Creatinine, mg/dL	0.29171	1.339
Albumin, g/dL	−0.27440	0.760
Uric acid, mg/dL	−0.02154	0.979

Term	Coefficient	Odds ratio per unit
Glucose, mg/dL	−0.00101	0.999
Total protein, g/dL	−0.20078	0.818
Cholesterol, mg/dL	0.00119	1.001
25(OH) vitamin D, nmol/L	−0.00539	0.995
Female sex	−0.02615	0.974
Diabetes	0.18337	1.201
Arthritis	−0.24444	0.783
Stroke	0.53830	1.713
Coronary heart disease	0.31449	1.370
Cancer	−0.38293	0.682
Former smoker	−0.05355	0.948
Current smoker	0.20477	1.227
Prior hip, wrist or spine fracture	0.93243	2.541
Parental hip fracture	−0.14696	0.863
Ever daily glucocorticoid use	0.14705	1.158
Postmenopausal	0.99824	2.714
Ever used female hormones	−0.79702	0.451
Alcohol ≥3 drinks/day	−0.33579	0.715
Mexican American	−0.77710	0.460
Other Hispanic	−0.36512	0.694
Non-Hispanic Black	−0.80116	0.449
Other/multiracial (incl. Asian)	−0.26758	0.765

Table S26: Predictors selected on the development set by each feature-selection method (used only for the re-implementation of the published approach).

Method	Selected predictors
Boruta (10)	age, PIR, BMI, weight, height, waist, alkaline phosphatase, creatinine, female sex, postmenopausal
Lasso (34)	all candidates except weight
XGBoost (10)	weight, postmenopausal, age, prior fracture, hormone use, female sex, coronary heart disease, non-Hispanic Black, Mexican American, BMI

Table S27: Full model versus OST by sex in the temporal validation set. AUROC difference with paired bootstrap 95% CI. OST cut-off columns show sensitivity / specificity / proportion referred (no fitting). Within each sex, the three-variable model had an AUROC within 0.001 of OST.

Sex	n	Cases	Full model AUROC	OST AUROC	Δ (95% CI)	OST < 2	OST < −1
Women	858	72	0.853	0.834	0.019 (−0.005 to 0.046)	0.875 / 0.552 / 48.4%	0.694 / 0.847 / 19.8%
Men	1,150	24	0.820	0.800	0.020 (−0.031 to 0.068)	0.833 / 0.689 / 32.2%	0.500 / 0.896 / 11.2%

Table S28: Predictor importance: mean absolute SHAP value of the full model in 500 development-set participants (probability scale, before recalibration).

Rank	Predictor	Mean |SHAP|
1	BMI, kg/m^2	0.0237
2	Weight, kg	0.0188
3	Postmenopausal	0.0176
4	Age, years	0.0174
5	Height, cm	0.0135
6	Waist, cm	0.0117
7	Ever used female hormones	0.0084
8	Non-Hispanic Black	0.0069
9	Income-to-poverty ratio	0.0067
10	Prior hip, wrist or spine fracture	0.0066
11	Mexican American	0.0062
12	Alkaline phosphatase, U/L	0.0054
13	Arthritis	0.0042
14	Cancer	0.0040
15	25(OH) vitamin D, nmol/L	0.0038
16	Stroke	0.0030
17	Total protein, g/dL	0.0027
18	Creatinine, mg/dL	0.0023
19	Current smoker	0.0023
20	Albumin, g/dL	0.0021
21	Diabetes	0.0020
22	Other Hispanic	0.0019
23	Coronary heart disease	0.0016
24	Education level (1–5)	0.0015
25	Other/multiracial (incl. Asian)	0.0014
26	Cholesterol, mg/dL	0.0014
27	Uric acid, mg/dL	0.0009
28	Former smoker	0.0008
29	Parental hip fracture	0.0008
30	Glucose, mg/dL	0.0007
31	Ever daily glucocorticoid use	0.0006
32	Female sex	0.0005
33	Alcohol $\geq$3 drinks/day	0.0005
34	Calcium, mg/dL	0.0001
35	Phosphorus, mg/dL	0.0001

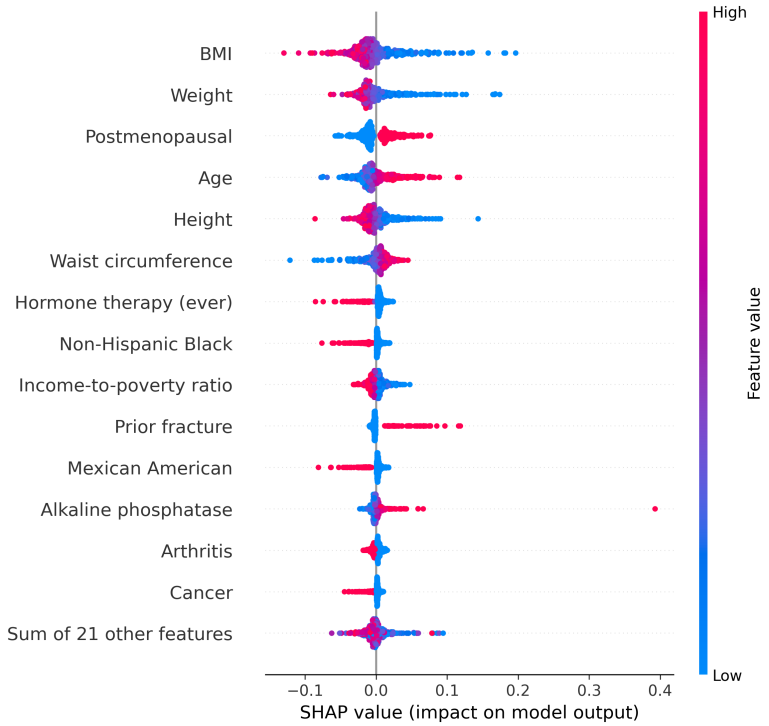


Figure S2: SHAP beeswarm summary for the full model in 500 development-set participants (the sample used for importance ranking). Colour indicates the predictor value; horizontal position indicates its effect on predicted risk (probability scale, before recalibration).

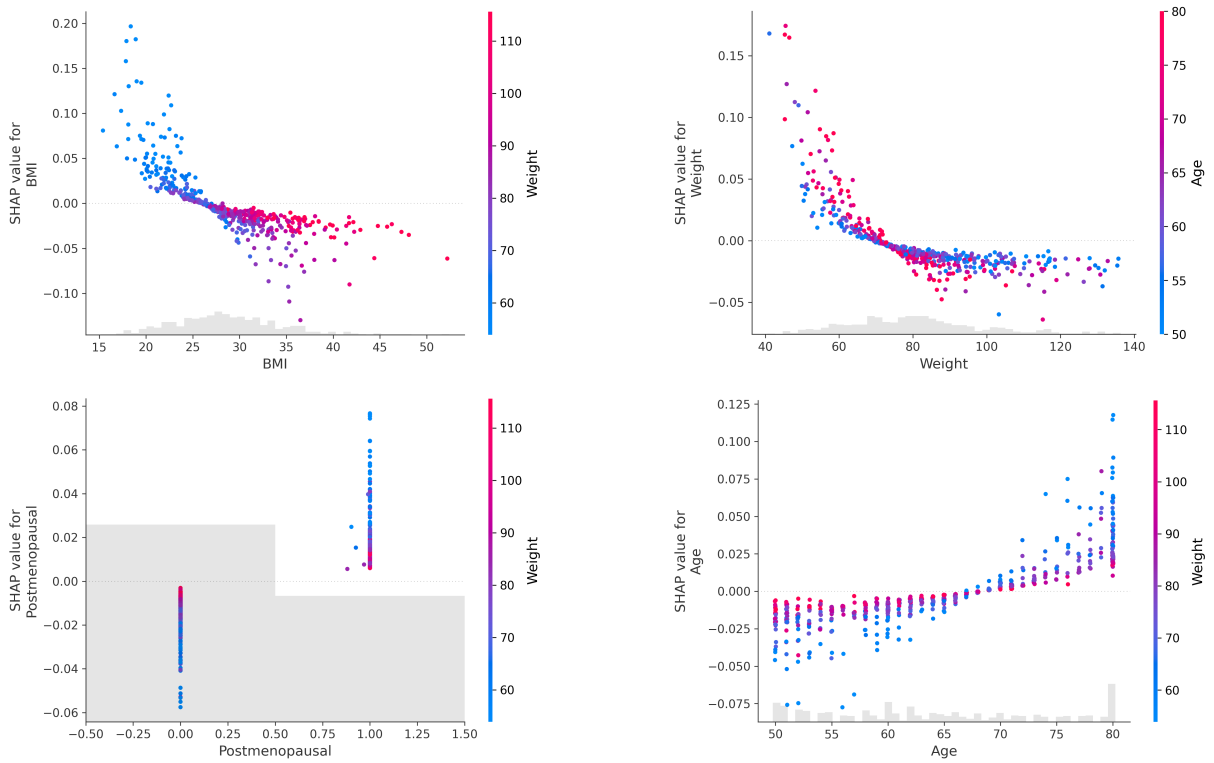


Figure S3: SHAP dependence plots for the four most important predictors (BMI, weight, postmenopausal status, age), full model, 500 development-set participants.

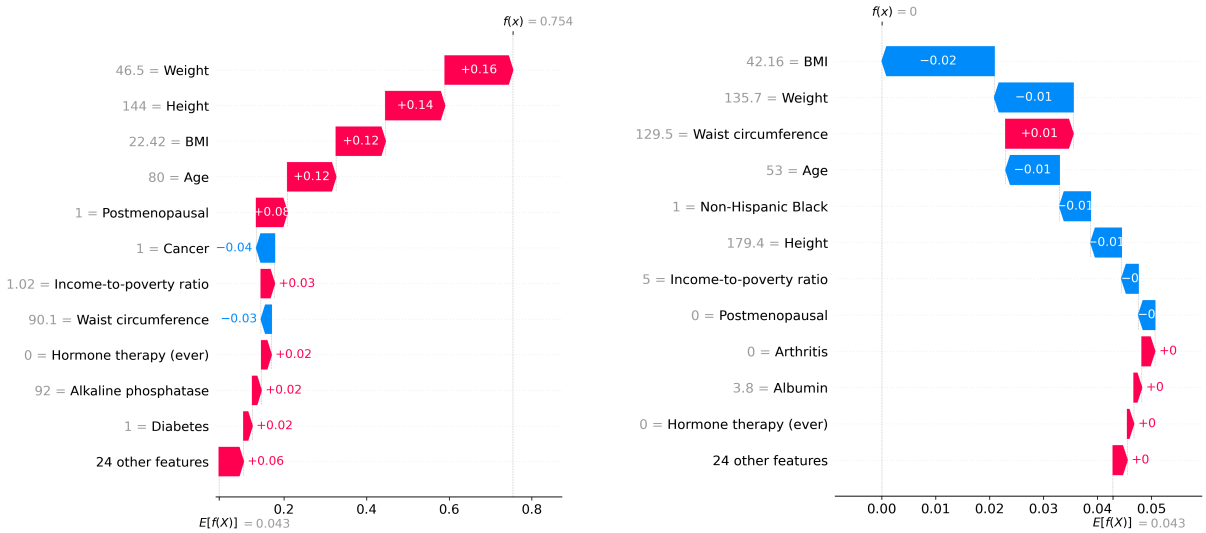


Figure S4: SHAP waterfall plots for the highest-risk (left) and lowest-risk (right) individuals in the explained development sample.

Table S29: Short-form equation (penalised logistic regression, recalibrated). Predicted probability $= 1/(1+e^{-\eta})$, where η is the intercept plus the sum of each coefficient multiplied by the predictor value. Units: weight in kg, age in years (capped at 80), height and waist in cm; binary predictors 1 = yes, 0 = no (postmenopausal and hormone use are 0 for men). Coefficients of correlated body-size measures are not individually interpretable. Development-set Youden threshold 4.0%.

Term	Coefficient	Odds ratio per unit
Intercept	−3.43385	
Weight	−0.09585	0.909
Postmenopausal	0.97643	2.655
Age	0.07253	1.075
Height	−0.00161	0.998
Waist	0.02118	1.021
Ever used female hormones	−0.83012	0.436
Prior hip, wrist or spine fracture	1.08190	2.950
Arthritis	−0.22820	0.796
Cancer	−0.30003	0.741
Stroke	0.69766	2.009

Table S30: Short form by sex in the temporal validation set: AUROC of the short form, the full model, the three-variable model and OST, and the short form's sensitivity and proportion referred at its development threshold (4.0%).

Sex	n	Cases	Short form	Full model	Age, weight, sex	OST	Sensitivity	Referred
Women	858	72	0.835	0.853	0.834	0.834	0.889	43.6%
Men	1,150	24	0.800	0.820	0.800	0.800	0.583	15.7%